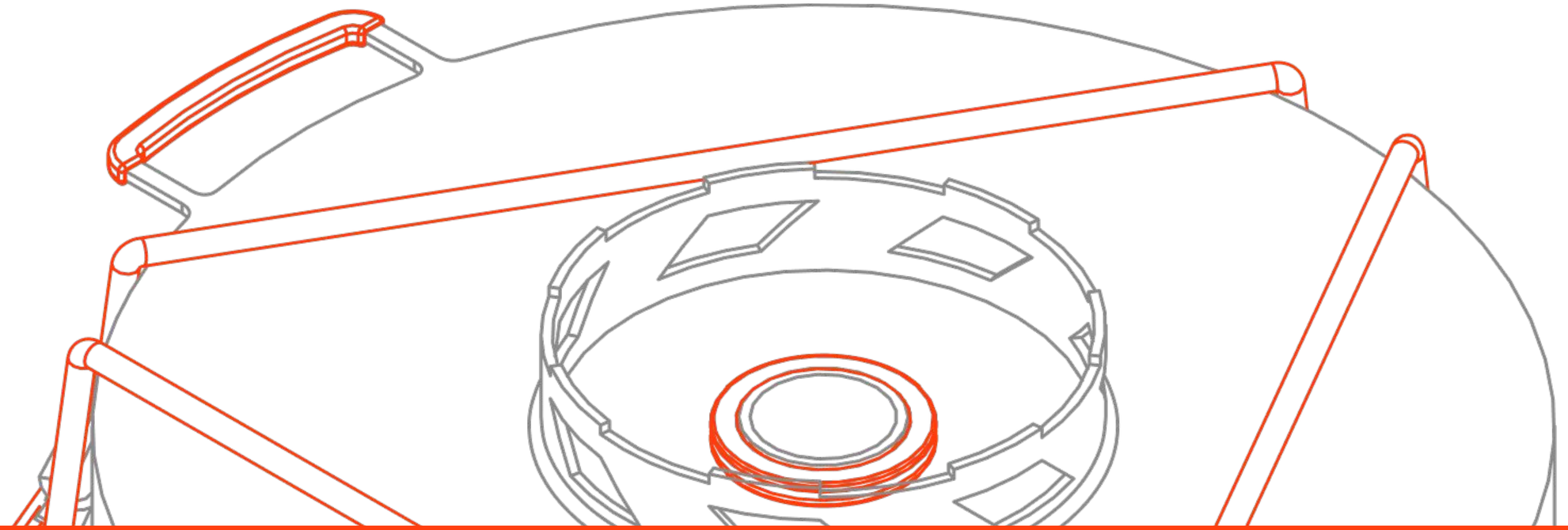
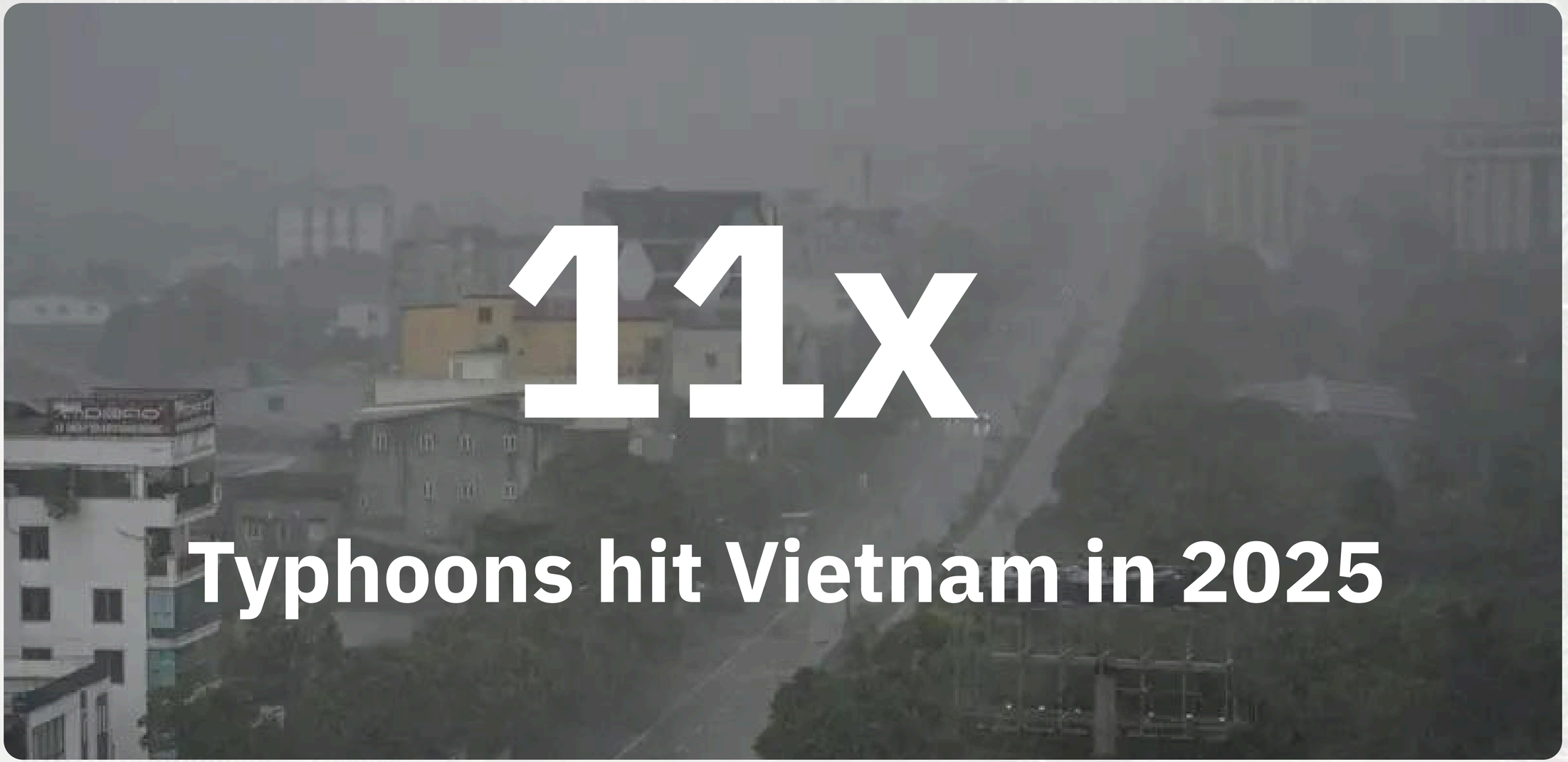


COOKING AND HEATING WITH WET WOOD POST-FLOODING IN RURAL VIETNAM

Student name: Francesco Rastelli
CID: 02310086
Email: fr422@ic.ac.uk
Supervisor: Audrey Gaulard



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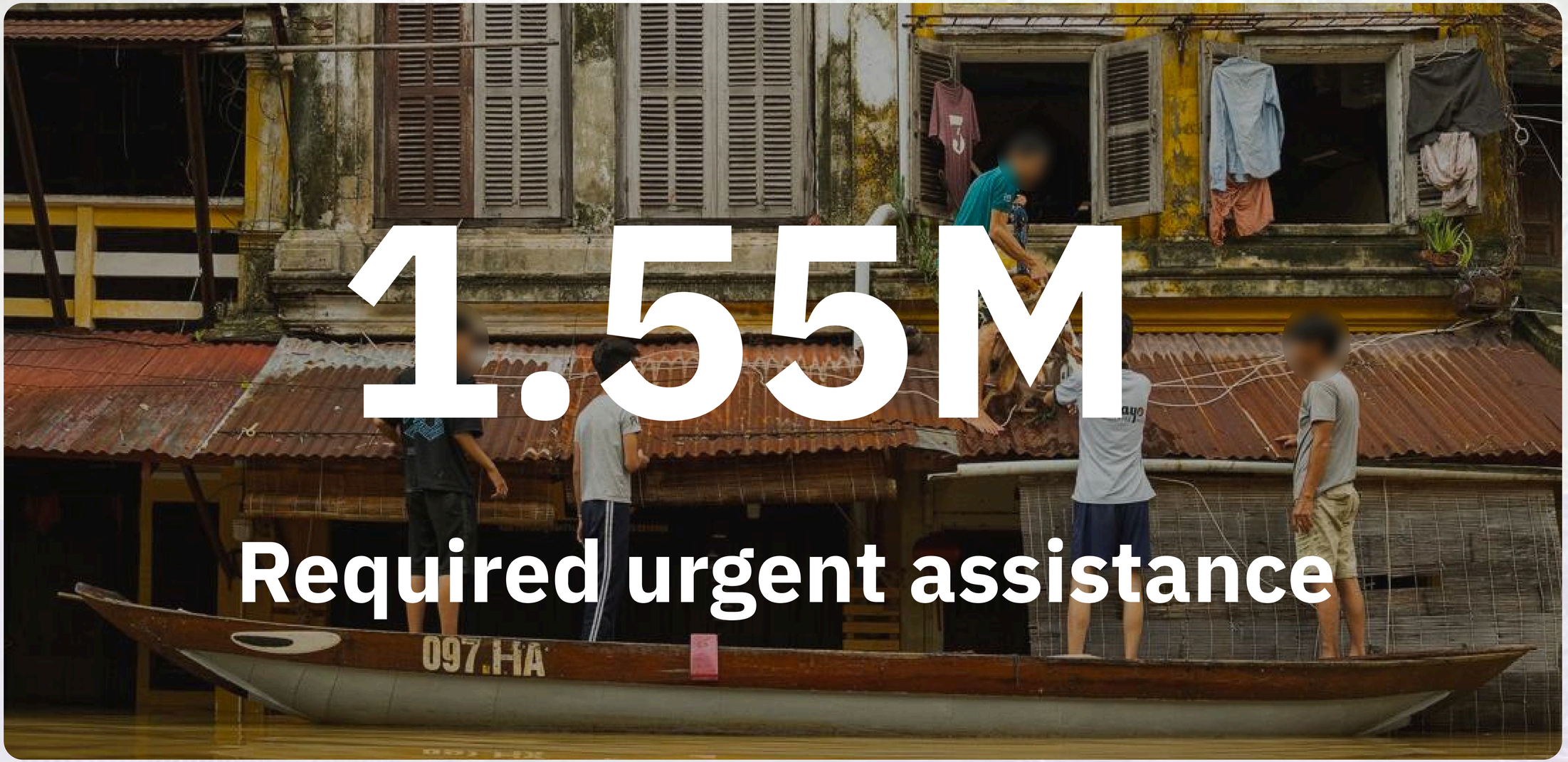

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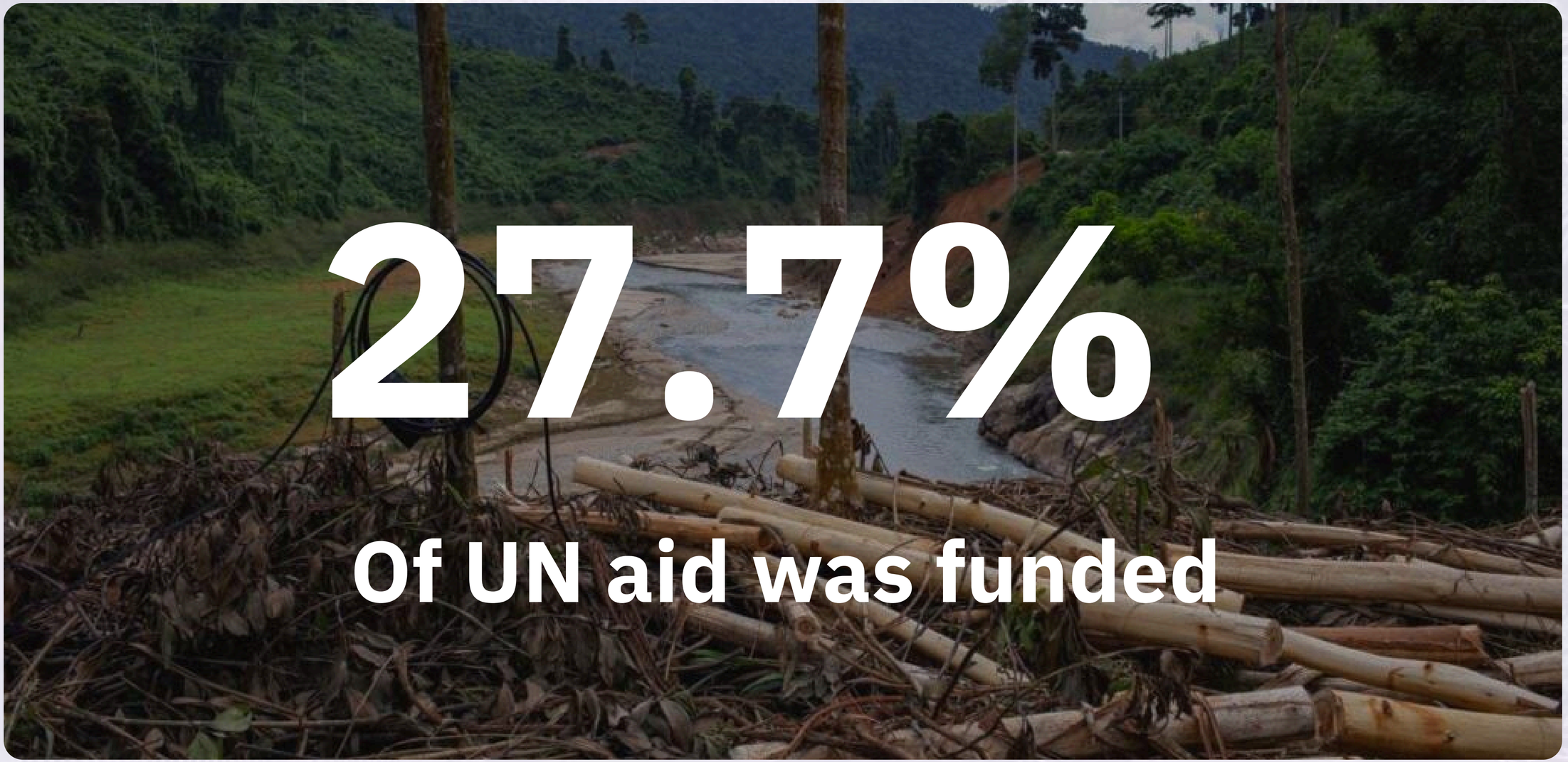
376,362

Victims reached by the UN

• Trang Anh (2025) Hanoians contribute each box of noodles, each truck to support people in flooded areas. https://news.laodong.vn/nguoi-viet-tu-te/nguoi-ha-noi-gop-tung-thung-mi-tung-chuyen-xe-hang-tiep-suc-dong-bao-vung-lu-1616089.ldo [Accessed 12th June 2026]



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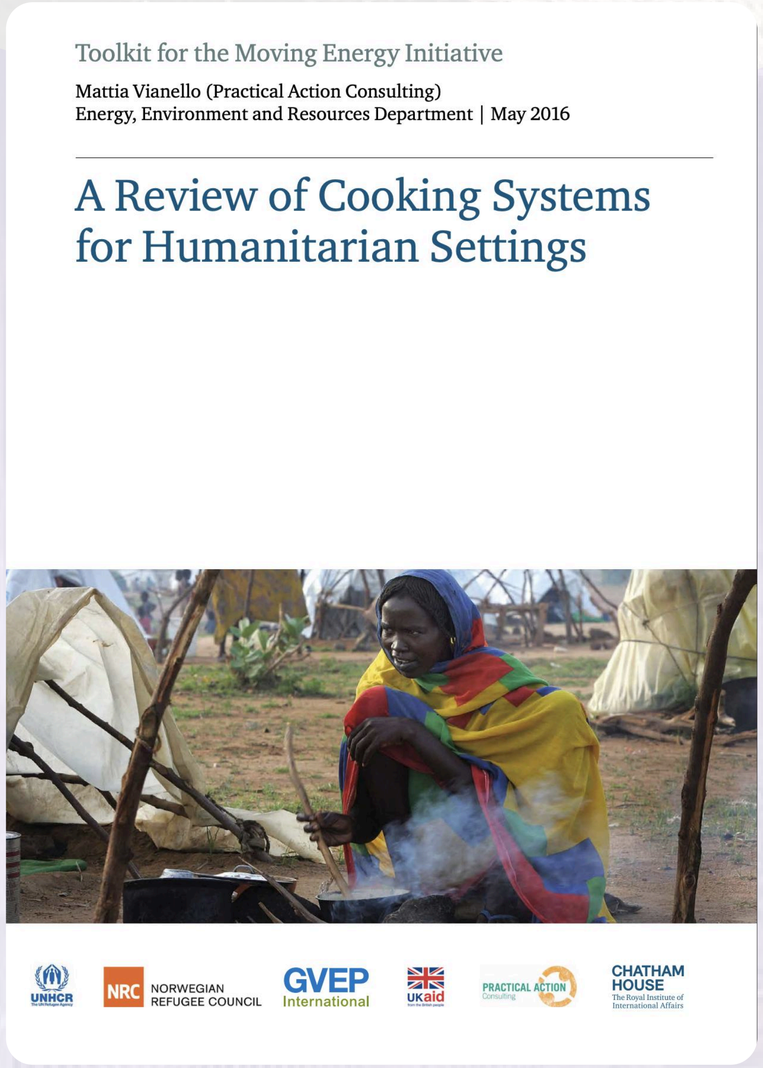
HOW CAN I MAKE REMOTE RURAL COMMUNITIES INDEPENDENTLY RESILIENT TO FLOODING WITH RESPECT TO FIREWOOD AND FUEL?

Improved cookstoves

Wood gasifiers

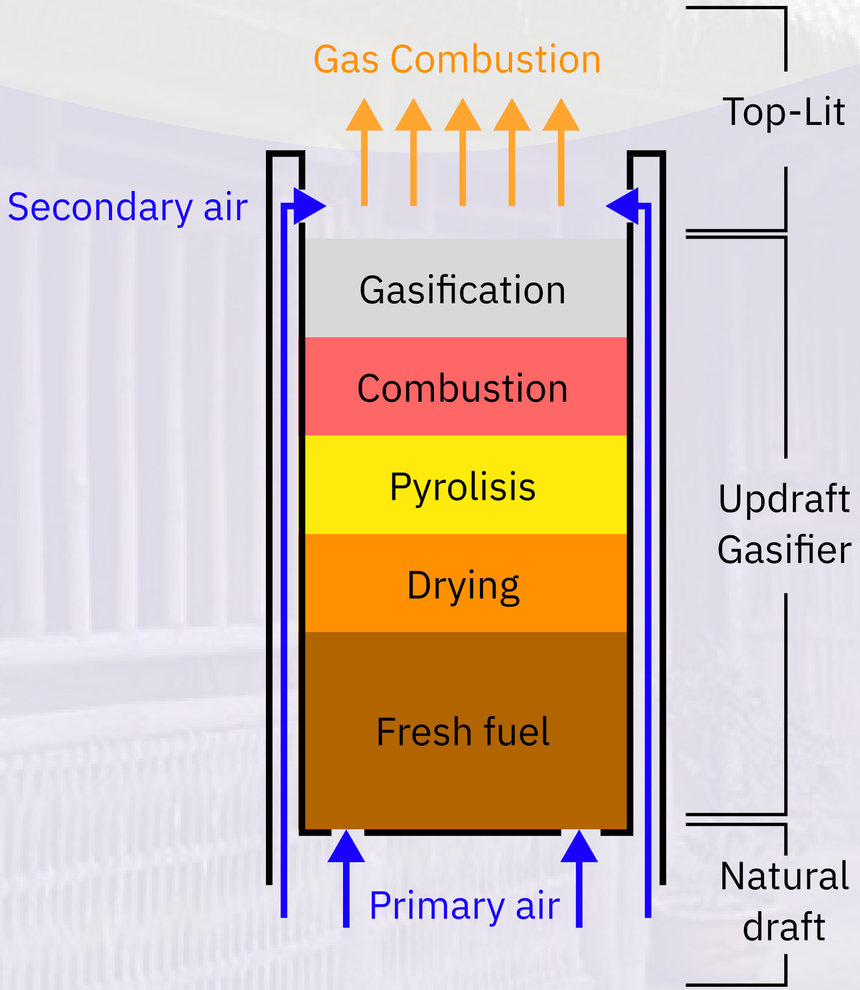
- Republic of Rwanda (n.d.) LDCF3 Project's Improved Cookstoves: A Dual Solution for Climate Change and Human Health https://www.rema.gov.rw/updates/news-details/ldcf3-projects-improved-cookstoves-a-dual-solution-for-climate-change-and-human-health [Accessed 12th June 2026]
- Randomonium (2020) Using Wood to Fuel a Generator! (How to Build a Wood Gasifier W/Demonstration) https://www.instructables.com/How-a-Wood-Gasifier-Works-and-How-I-Built-Mine-Out/ [Accessed 12th June 2026]



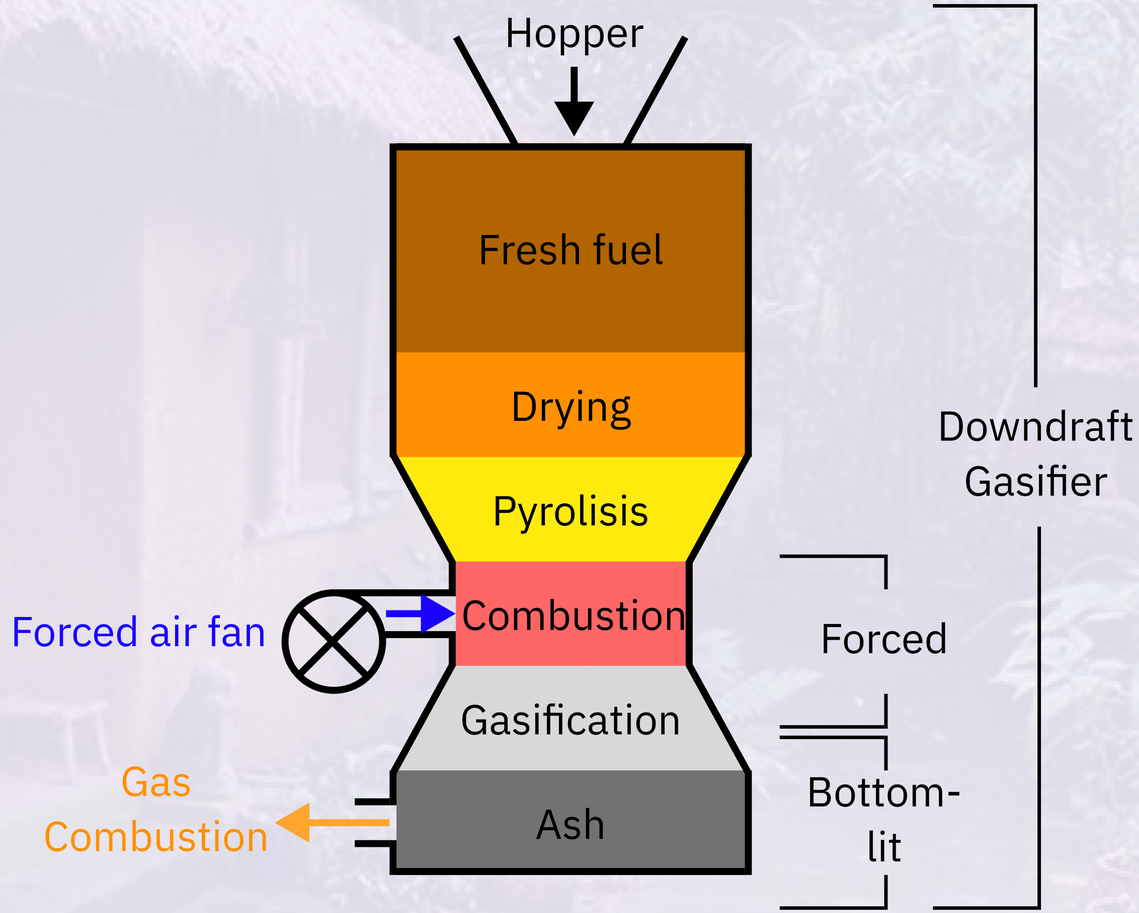
1. Durability
2. Stability
3. Ease of lighting
4. Cooking speed
5. Fuel type
6. Portability
7. Radiant heat output
8. Fit with existing utensils
9. Visual attractiveness and build quality
10. Price and availability
11. Clean burning performance

• Vianello M. (2016) *A Review of Cooking Systems for Humanitarian Settings.* Practical Action Consulting. https://www.chathamhouse.org/sites/default/files/publications/research/2016-05-19-mei-review-of-cooking-systems-vianello.pdf

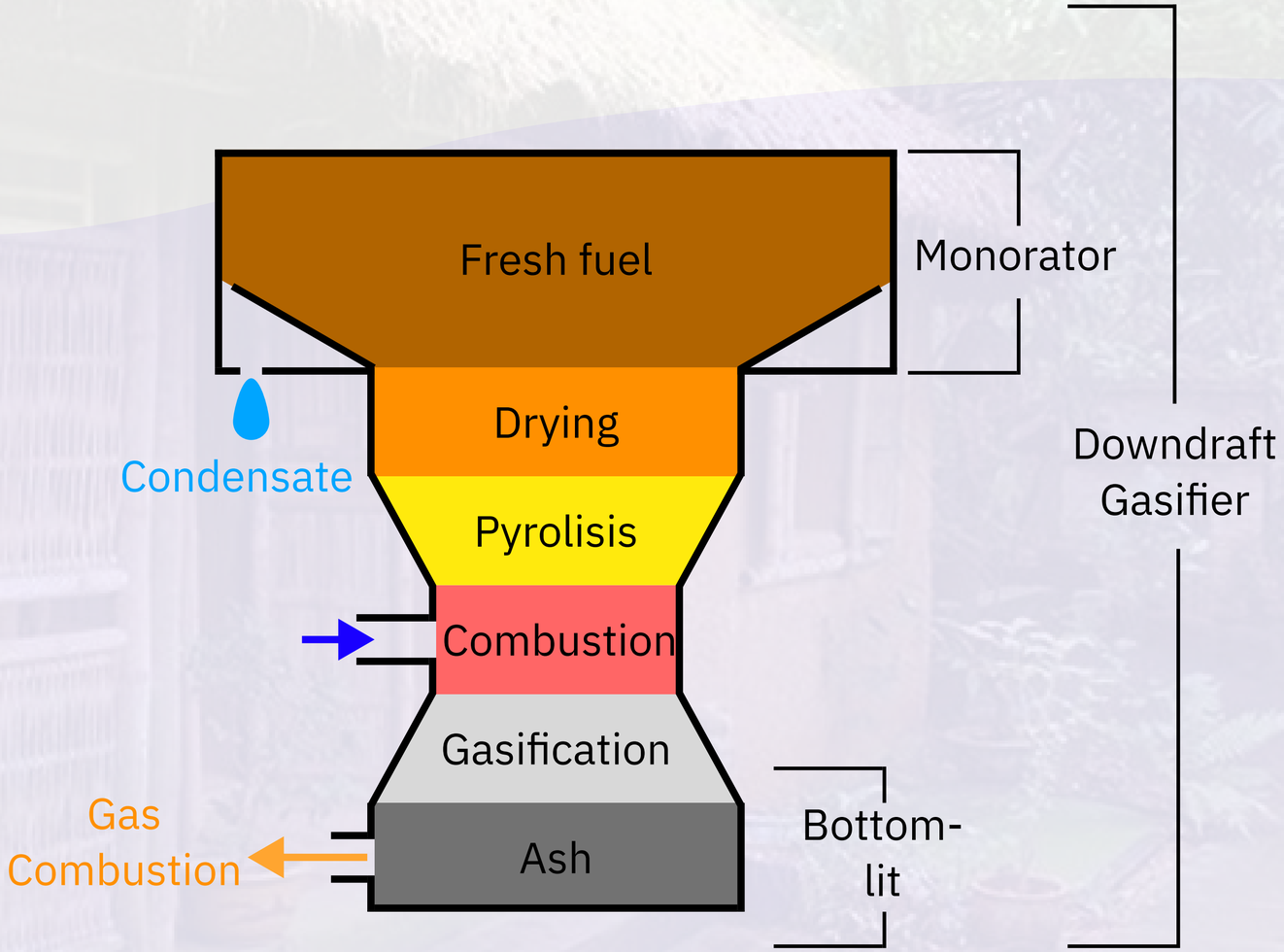
Top-lit Natural Updraft Gasifier



Bottom-lit Forced Downdraft Gasifier



Bottom-lit Monorator Downdraft Gasifier



Distribution should occur both proactively and reactively.

In rural Vietnam, marketing should be reinforced by respected figures within the community, such as health workers or teachers, and focus on immediate personal benefits such as saved money or time’ (Vianello, 2016) by avoiding the need to collect dry firewood.

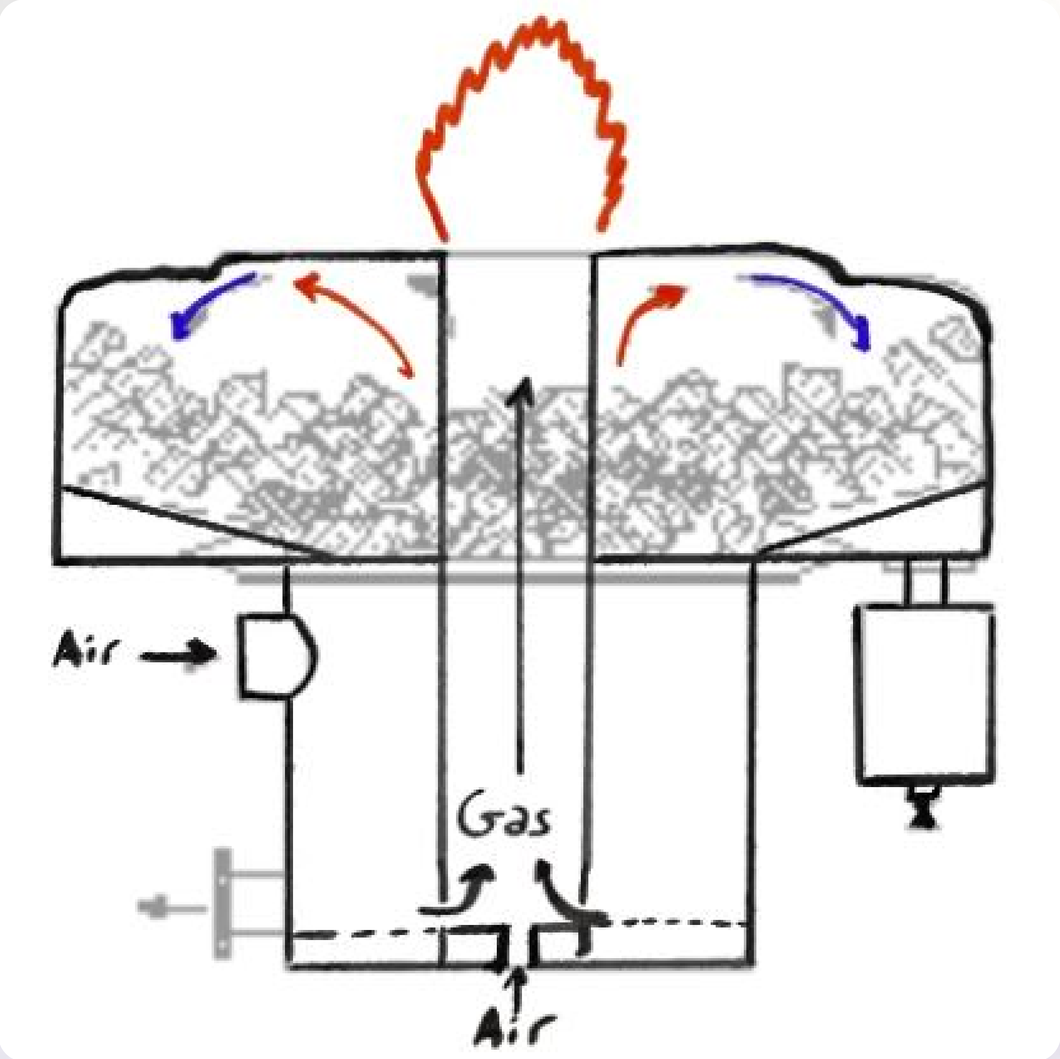
A mix of VLE and a Proprietary sales network would enable victims in remote rural villages to be reached and to exploit Cash & Voucher Assistance (CVA) received by the communities, facilitating easier financing for victims.

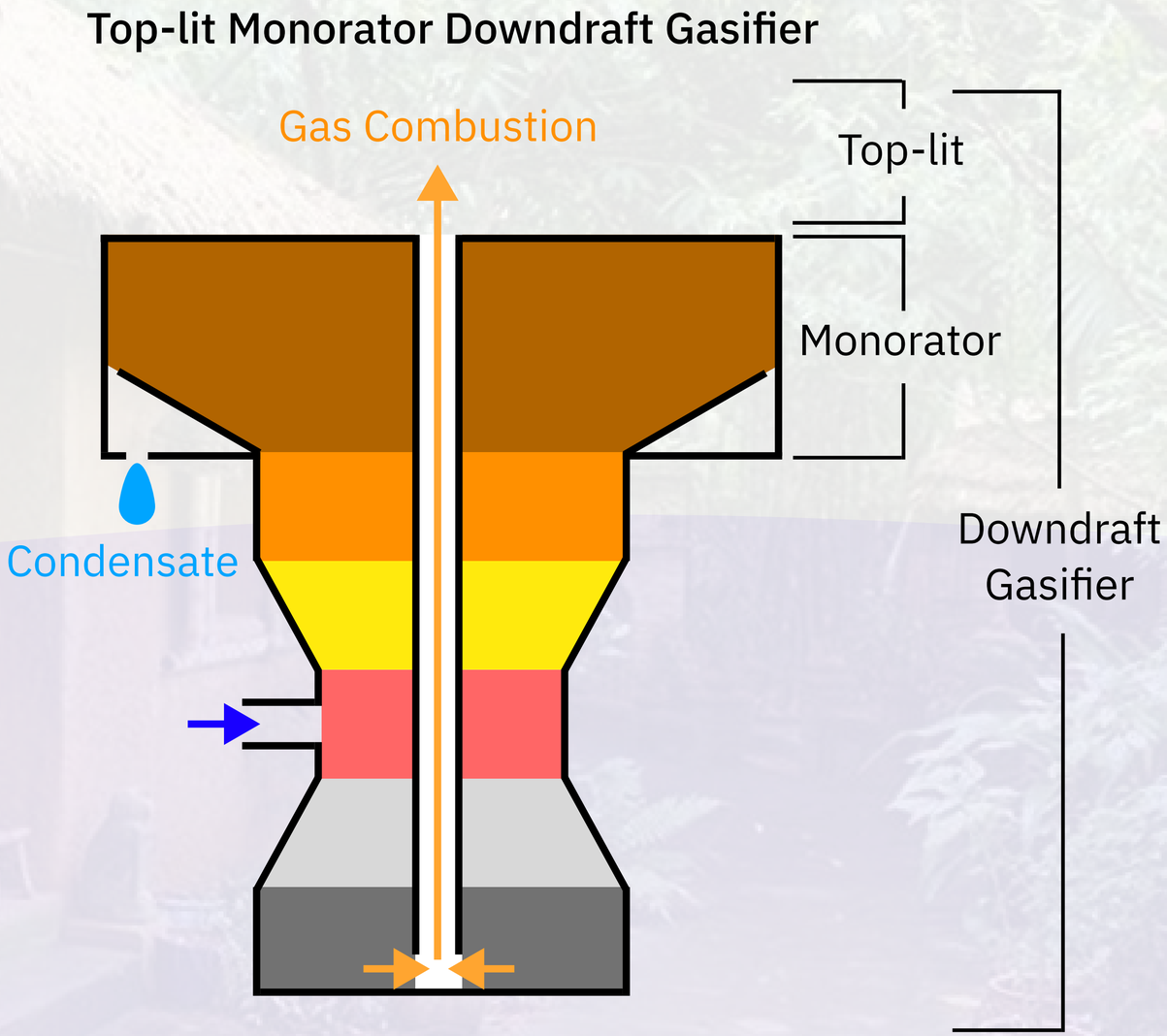
• Vianello M. (2016) *A Review of Cooking Systems for Humanitarian Settings.* Practical Action Consulting. https://www.chathamhouse.org/sites/default/files/publications/research/2016-05-19-mei-review-of-cooking-systems-vianello.pdf

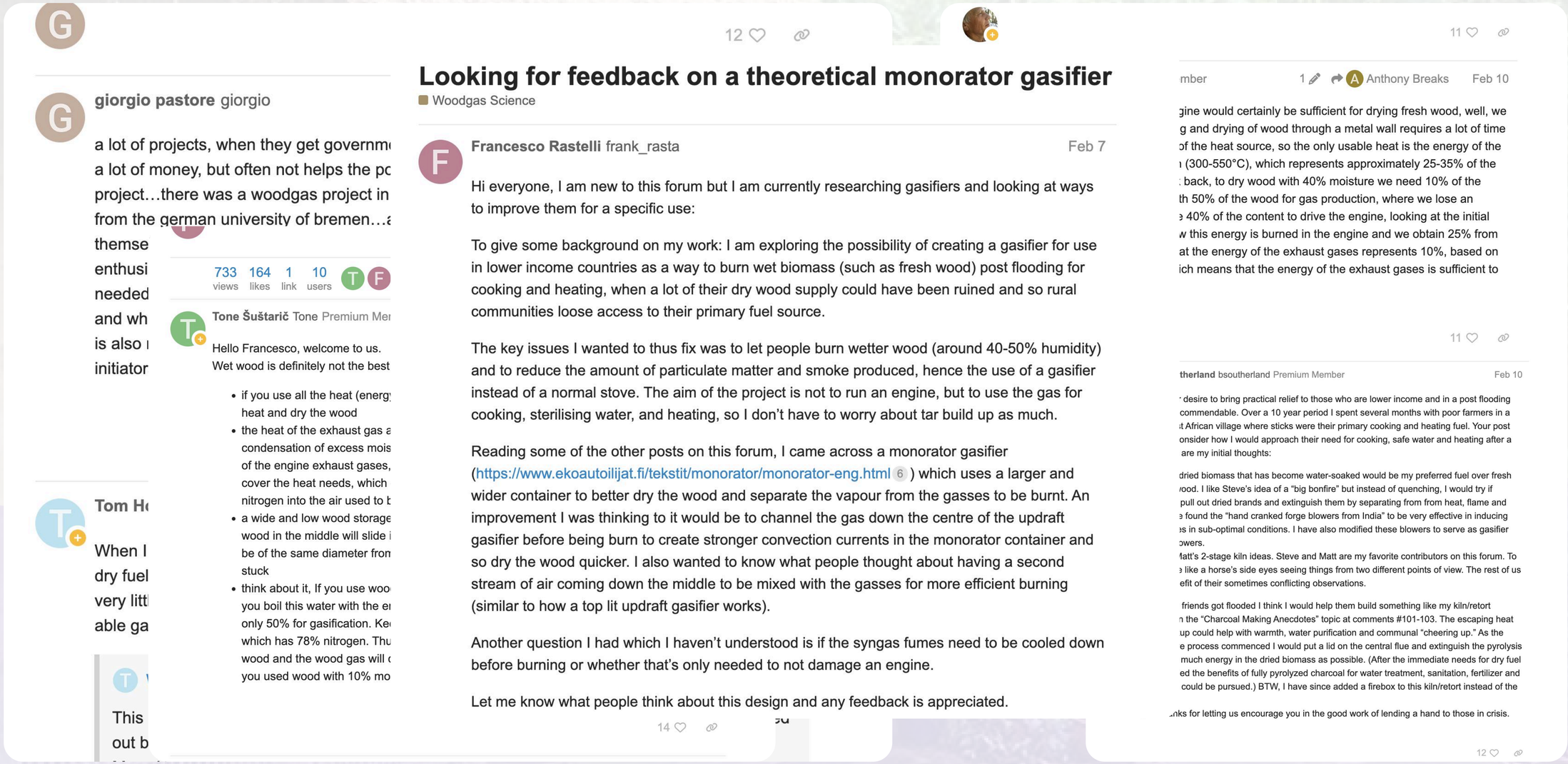
CAN A SMALL-SCALE WOOD GASIFIER STOVE BE USED FOR COOKING AND HEATING WITH WET WOOD POST-FLOOD IN VIETNAM?

Additional constraints:

1. 450mm x 450mm x 450mm
2. 20kg max weight
3. Water resistant design
4. Water resistant materials
5. Natural draft
6. Insulated
7. Bed height should be 0.5 times the diameter


Air
Gas
Air


Top-lit Monorator Downdraft Gasifier
Gas Combustion
Top-lit
Monorator
Condensate
Downdraft
Gasifier



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Looking for feedback on a theoretical monorator gasifier

Woodgas Science

Francesco Rastelli frank_rasta — Feb 7

Hi everyone, I am new to this forum but I am currently researching gasifiers and looking at ways to improve them for a specific use:

To give some background on my work: I am exploring the possibility of creating a gasifier for use in lower income countries as a way to burn wet biomass (such as fresh wood) post flooding for cooking and heating, when a lot of their dry wood supply could have been ruined and so rural communities loose access to their primary fuel source.

The key issues I wanted to thus fix was to let people burn wetter wood (around 40-50% humidity) and to reduce the amount of particulate matter and smoke produced, hence the use of a gasifier instead of a normal stove. The aim of the project is not to run an engine, but to use the gas for cooking, sterilising water, and heating, so I don't have to worry about tar build up as much.

Reading some of the other posts on this forum, I came across a monorator gasifier (https://www.ekoautoilijat.fi/tekstit/monorator/monorator-eng.html 6) which uses a larger and wider container to better dry the wood and separate the vapour from the gasses to be burnt. An improvement I was thinking to it would be to channel the gas down the centre of the updraft gasifier before being burn to create stronger convection currents in the monorator container and so dry the wood quicker. I also wanted to know what people thought about having a second stream of air coming down the middle to be mixed with the gasses for more efficient burning (similar to how a top lit updraft gasifier works).

Another question I had which I haven't understood is if the syngas fumes need to be cooled down before burning or whether that's only needed to not damage an engine.

Let me know what people think about this design and any feedback is appreciated.

14

giorgio pastore giorgio

a lot of projects, when they get governm
a lot of money, but often not helps the pc
project…there was a woodgas project in
from the german university of bremen…a
themse
enthusi
needed
and wh
is also
initiator

733 views 164 likes 1 link 10 users

Tone Šuštarič Tone Premium Mer

Hello Francesco, welcome to us.
Wet wood is definitely not the best

- if you use all the heat (energ
heat and dry the wood
- the heat of the exhaust gas a
condensation of excess mois
of the engine exhaust gases,
cover the heat needs, which
nitrogen into the air used to b
- a wide and low wood storage
wood in the middle will slide i
be of the same diameter fron
stuck
- think about it, If you use woo
you boil this water with the e
only 50% for gasification. Ke
which has 78% nitrogen. Thu
wood and the wood gas will
you used wood with 10% mo

Tom H

When I
dry fuel
very litt
able ga

This
out b

11

mber 1 Anthony Breaks Feb 10

gine would certainly be sufficient for drying fresh wood, well, we
g and drying of wood through a metal wall requires a lot of time
of the heat source, so the only usable heat is the energy of the
(300-550°C), which represents approximately 25-35% of the
back, to dry wood with 40% moisture we need 10% of the
th 50% of the wood for gas production, where we lose an
40% of the content to drive the engine, looking at the initial
this energy is burned in the engine and we obtain 25% from
at the energy of the exhaust gases represents 10%, based on
ich means that the energy of the exhaust gases is sufficient to

11

therland bsoutherland Premium Member Feb 10

desire to bring practical relief to those who are lower income and in a post flooding
commendable. Over a 10 year period I spent several months with poor farmers in a
t African village where sticks were their primary cooking and heating fuel. Your post
onsider how I would approach their need for cooking, safe water and heating after a
are my initial thoughts:

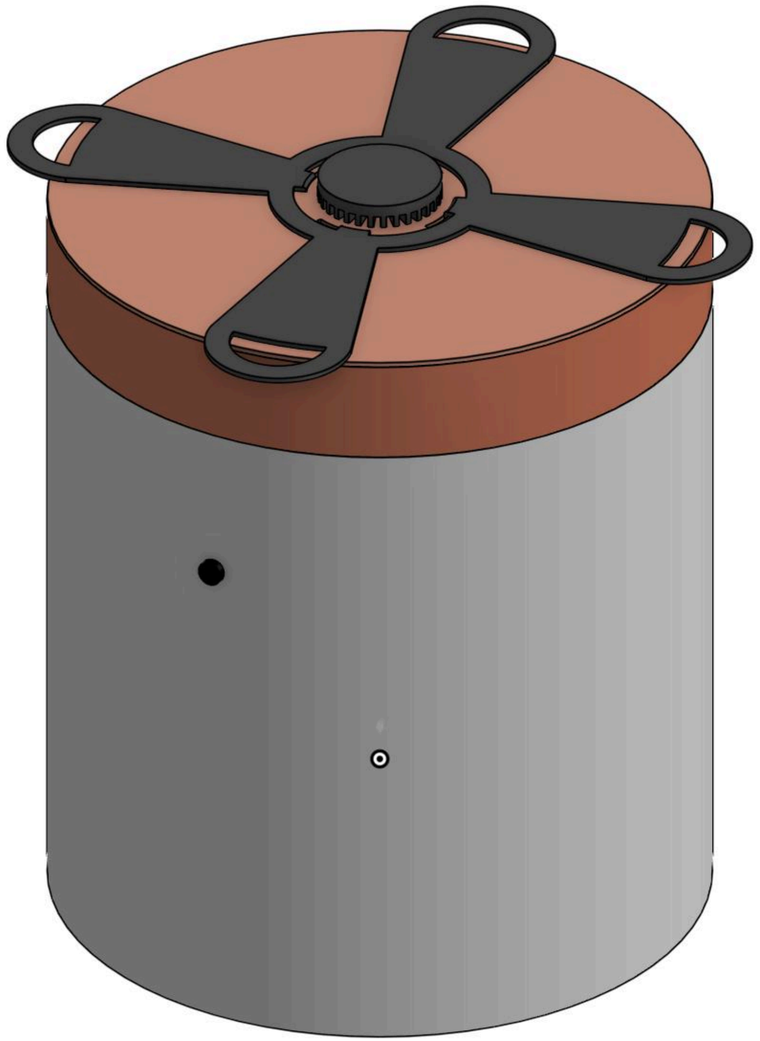
dried biomass that has become water-soaked would be my preferred fuel over fresh
/ood. I like Steve's idea of a "big bonfire" but instead of quenching, I would try if
pull out dried brands and extinguish them by separating from from heat, flame and
found the "hand cranked forge blowers from India" to be very effective in inducing
s in sub-optimal conditions. I have also modified these blowers to serve as gasifier
owers.
latt's 2-stage kiln ideas. Steve and Matt are my favorite contributors on this forum. To
like a horse's side eyes seeing things from two different points of view. The rest of us
efit of their sometimes conflicting observations.

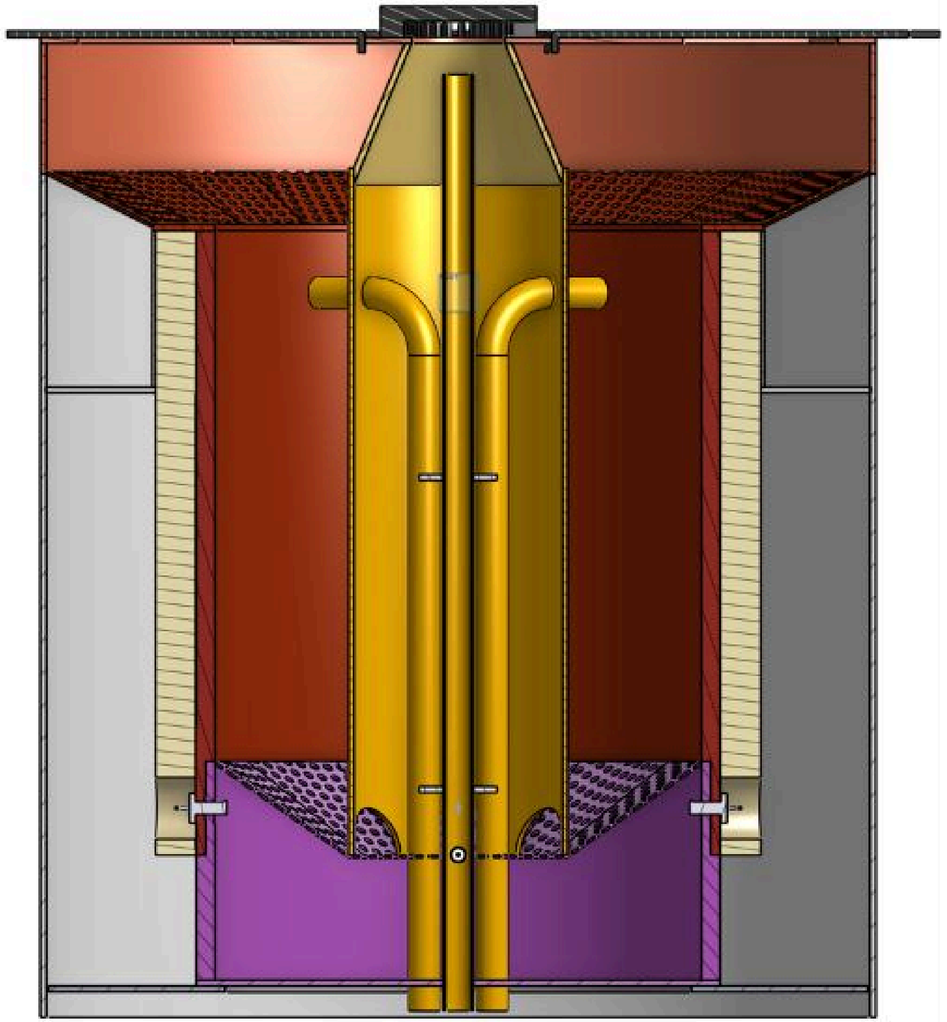
friends got flooded I think I would help them build something like my kiln/retort
n the "Charcoal Making Anecdotes" topic at comments #101-103. The escaping heat
up could help with warmth, water purification and communal "cheering up." As the
e process commenced I would put a lid on the central flue and extinguish the pyrolysis
much energy in the dried biomass as possible. (After the immediate needs for dry fuel
ed the benefits of fully pyrolyzed charcoal for water treatment, sanitation, fertilizer and
could be pursued.) BTW, I have since added a firebox to this kiln/retort instead of the

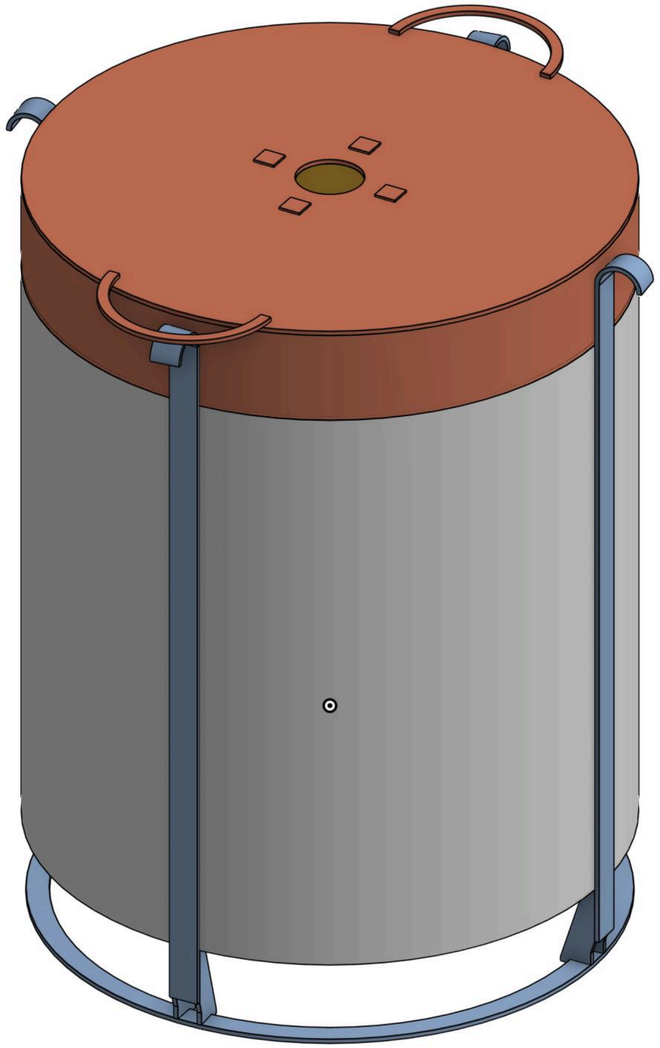
nks for letting us encourage you in the good work of lending a hand to those in crisis.

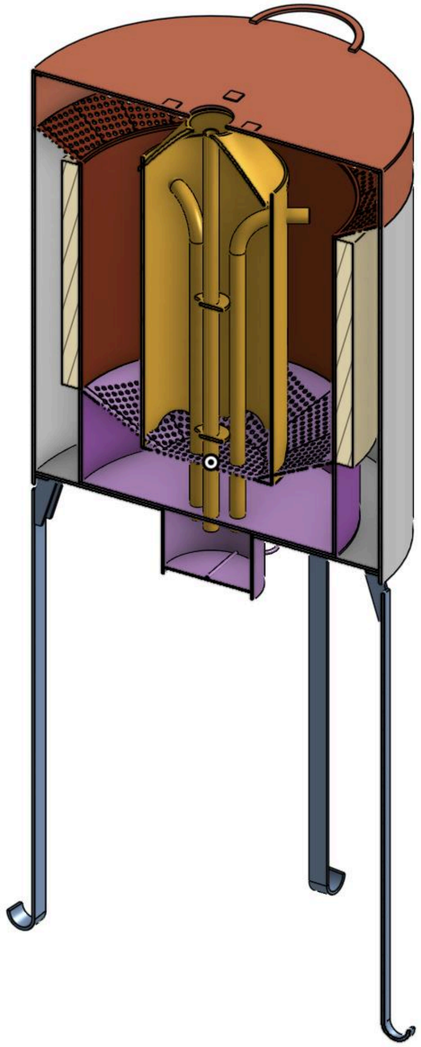
12

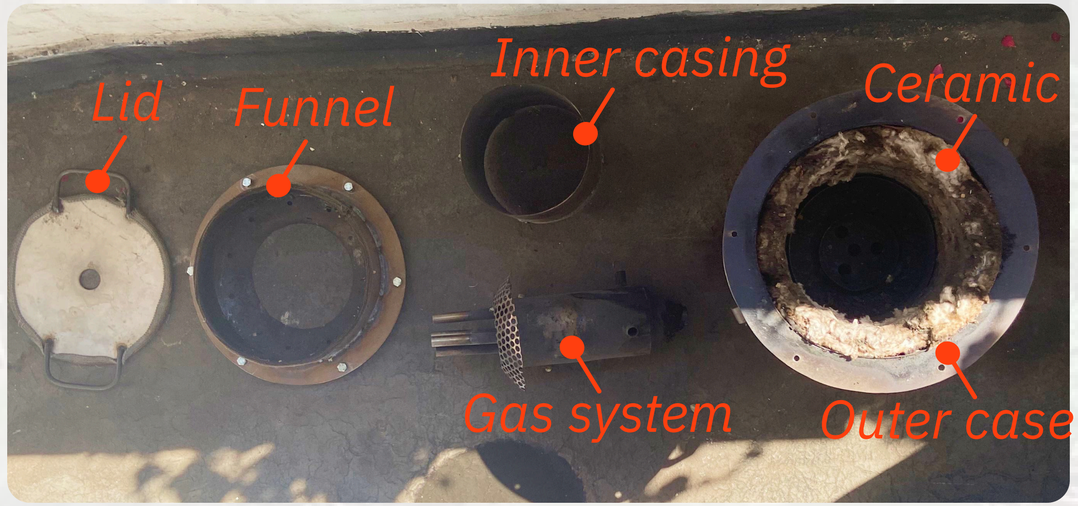
- Rastelli F. (2026) Discussion over the design of a theoretical monorator gasifier. Looking for feedback on a theoretical monorator gasifier. 7th February. https://forum.driveonwood.com/t/looking-for-feedback-on-a-theoretical-monorator-gasifier/8454. [Accessed 18th May 2026]










Lid
Funnel
Inner casing
Ceramic
Gas system
Outer case

1. Partially fill the cylinder with wood chips
2. Light them with the bottom valve fully open
3. Wait for them to get hot enough to charr
4. Filling the hopper up with the rest of the wood chips and seal the lid shut
5. The exhaust gasses can then be lit after a few minutes have passed

Test the heating and cooking of water and rice, duration of flame and temperatures across the device.

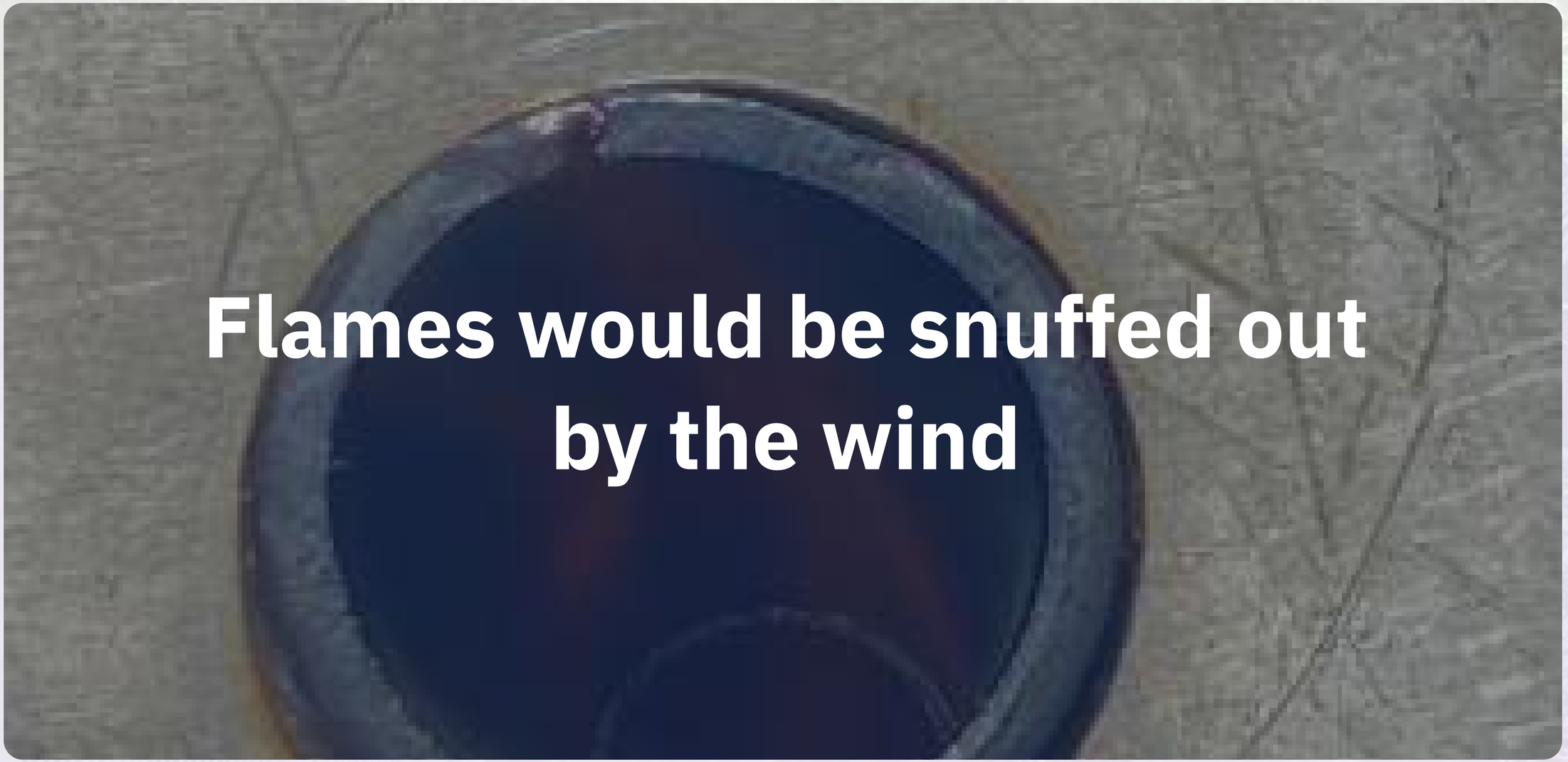
Real world testing of the device

First successful flame lasted
4 mins

Charred wood chips and ash
clogged the small holes

Syngas escaped the lid due to a bad seal

The stove required a forced draft to run


Flames would be snuffed out
by the wind

Using fresh cut wood sticks resulted in a 12min flame

Wood drying at the top of the hopper

8h → 35mins ------- 12h → 50mins ------- 24h → 55mins.

1. The success will be evaluated by analysing the ability of the wood gasifier stove to boil water, sterilise it, and cook rice in it.
2. The maximum moisture content of wood that the wood gasifier stove can handle.
3. Portability of the stove so it can be transported to affected victims when roads are too damaged for cars, vans or other traditional large-scale modes of distribution.
4. The stove's ability to warm the surrounding area to heat victims.
5. How well the design meets the Moving Energy Initiative's 11 design considerations (Vianello, 2016).

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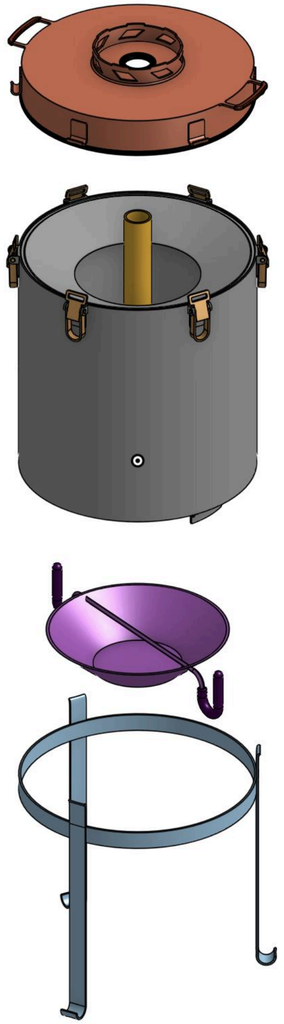
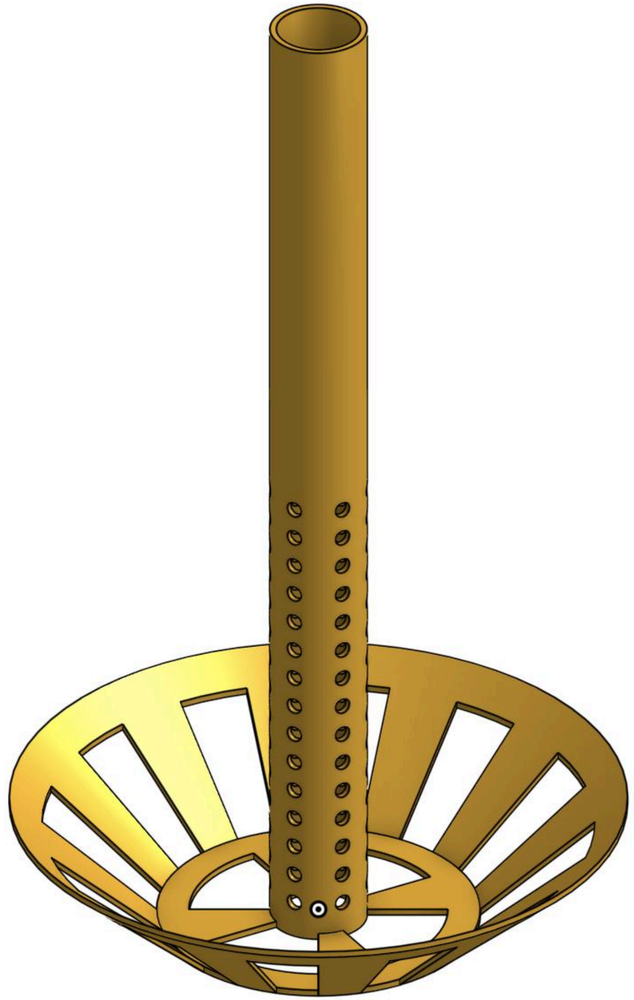
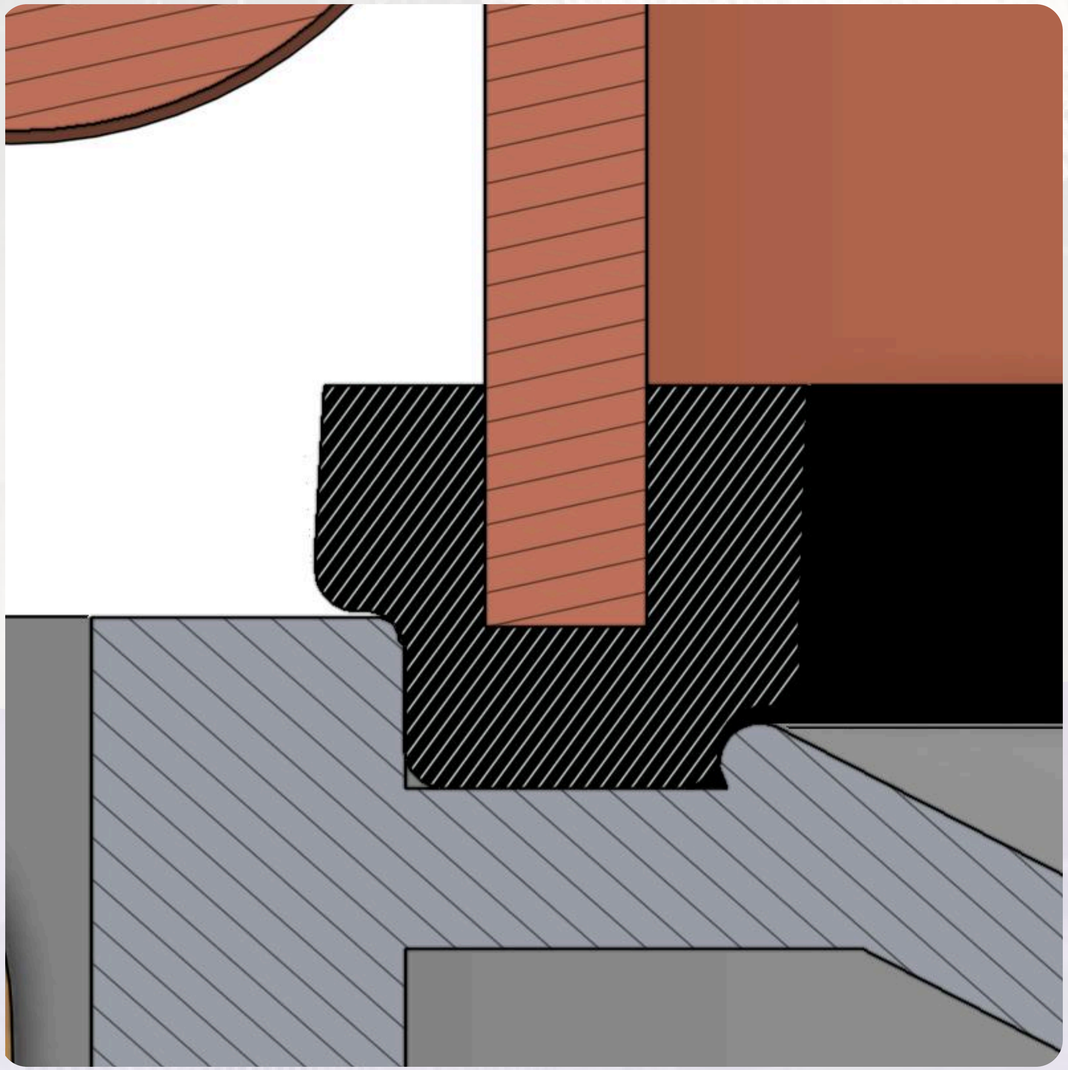
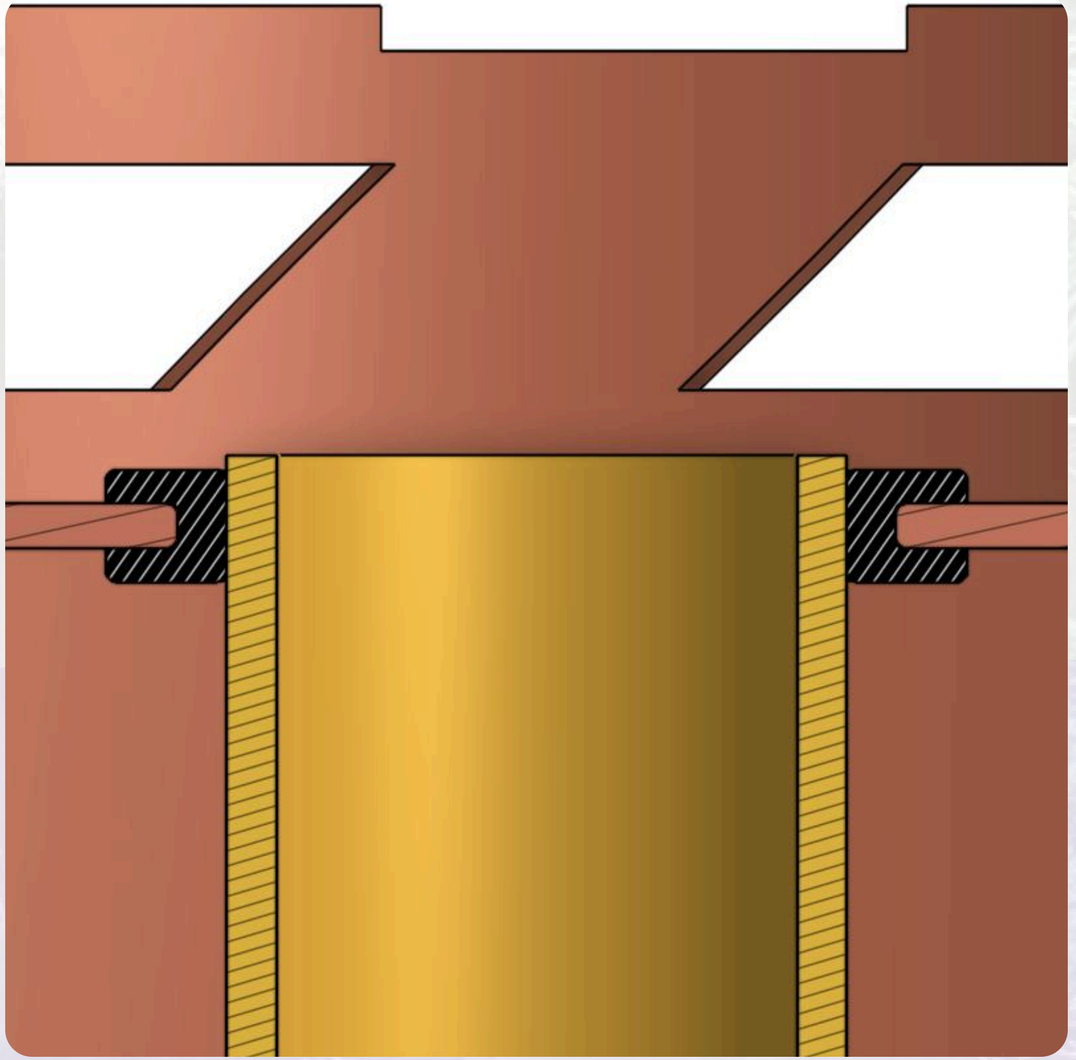
‘The inability of users to see the wood as it was burning would create distrust in the stove’ (Anonymous)

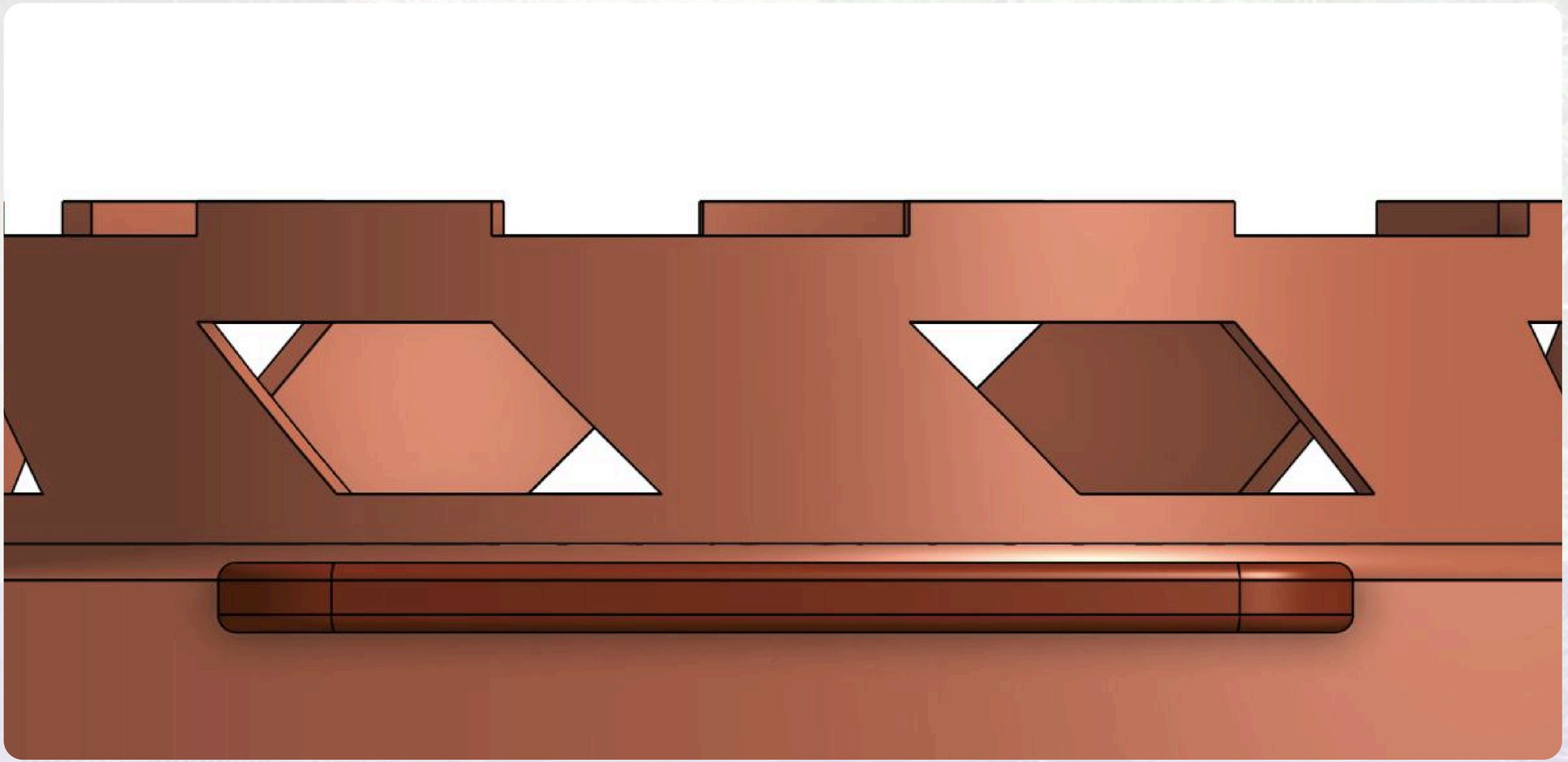
Information provided during a personal interview, May 15th, 2026. The interviewee's name has been withheld by request to protect their anonymity.

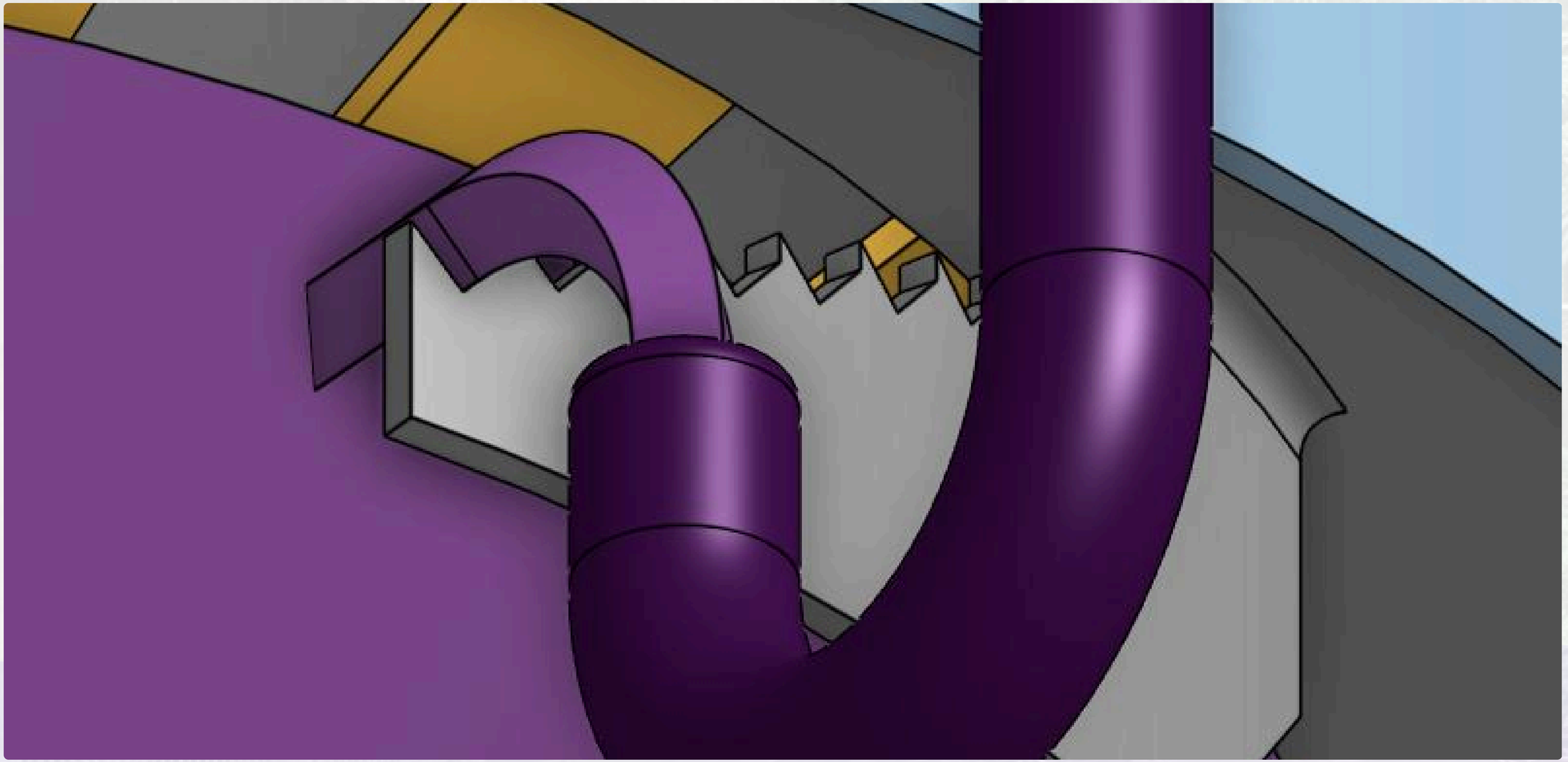






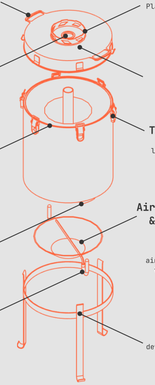






WETFIRE STOVE MANUAL EN

1 Components



Carrying handles
Use these to carry the device when the clamps are locked

Stove
Place pots on top of the lit flame

O-ring
Ensures a tight seal between the lid and the gas tube

Lid
Ensure clamps are closed to seal

Gasket
Ensures a tight seal between the lid and the body

Toggle clamps
Clamp to seal the lid shut, and carry the device around

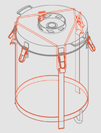
Ratchet system
Move the ash catcher up and down the ratchet to control airflow

Air flow valve & Ash catcher
Moving it up and down the ratchet system to control airflow. Remove when full to dispose of ash.

Valve handles
Use the silicone handles to protect your hands from the high heat

Stand
Use to lift the device off the floor and protect when packed away

2 Unpacking



(1) Remove the elastic straps
Remove the elastic straps which hold the stand to the body.

(2) Flip the stand
Flip the stand and place the device on top of it, ensuring it's stable.

(3) Unclasp the clamps
Undo the 6 clamps holding the lid on and remove the lid for loading.

3 Usage

(1) Pre-load and lighting
Remove the lid and load the hopper halfway with wood. Light the wood from below ensuring the valve is fully open.

(2) Full-load and sealing
When the wood is visibly charring, load the remainder of the wood up to the top of the hopper, wait 5 minutes, and then seal the lid.

(3) Lighting of gas
When mostly see-through gas begins to exit the central pipe, light it on fire with a lighter to turn on the stove.

(4) Running procedures
If too much thick smoke comes out, decrease the airflow. If too little gas is produced but not a lot of smoke, increase the airflow.

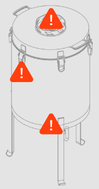
4 Technical Information

(1) Wood reccomendations
The device can be fuelled by dry, fresh cut or wet wood chips or sticks. Recommended dimensions are between 20mm and 60mm in size (0.8" to 2.4").

While the device is designed to be used with wet wood, it is recommended to fist light it in stage 1 with dryer wood for a quicker start up.

(2) Cleaning and maintenance
Cleaning:
(A) Remove any left over wood chips or large chunks inside the hopper.
(B) Shake the device to make any loose ash fall into the ash catcher.
(C) Dispose of the ash collected below into compost or general waste.

Maintenance:
(A) The gasket and O-ring can fail long term due to wear and tear. To replace them, remove the broken seal, clean the surface, and place a new gasket.
(B) Clamps are press-fit, so if they fail, new clamps can be installed easily.
(C) Silicone handles can be press-fitted on.



(3) Safety information
(!) The lid, stove top and bottom surface of the device can reach very high temperatures and burn the user.

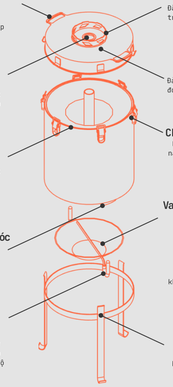
If moving or using the device, please use the silicone handles to reduce the chance of injury.

(!) Before cleaning the device, remove the lid after use and wait 2 hours to ensure the inner metal has cooled down enough.

(!) Do not use the device inside.

HƯỚNG DẪN SỬ DỤNG BẾP LÒ ĐỐT VN

1 Linh kiện



Tay cầm xách
Sử dụng những phụ kiện này để giữ thiết bị khi các kẹp được khóa.

Cái lò
Đặt các chậu cây lên trên bánh flane đang cháy.

Nắp
Đảm bảo các kẹp được đóng kín để bịt kín.

Vòng chữ O
Đảm bảo độ kín khít giữa nắp và ống dẫn khí.

Chuyển đổi kẹp
Dùng kẹp để bịt kín nắp và mang thiết bị đi khắp nơi.

Vòng đệm
Đảm bảo độ kín khít giữa nắp và thân máy.

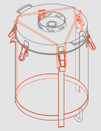
Van điều tiết lưu lượng khí & Bộ phận hứng tro
Di chuyển bộ phận lên xuống theo hệ thống bánh răng để điều chỉnh luồng khí. Tháo ra khi đầy để đổ tro.

Hệ thống bánh cóc
Di chuyển bộ phận hứng tro lên xuống theo cơ cấu bánh răng để điều chỉnh luồng không khí.

Đứng
Dùng để nâng thiết bị lên khỏi sàn và bảo vệ khi cất giữ.

Tay cầm van.
Hãy sử dụng tay cầm bằng silicon để bảo vệ tay khỏi nhiệt độ cao.

2 Giải nén



(1) Tháo dây thun
Tháo các dây chun giữ giá đỡ vào thân máy.

(2) Lật giá đỡ
Lật ngược giá đỡ và đặt thiết bị lên trên, đảm bảo thiết bị được đặt chắc chắn.

(3) Tháo các kẹp
Tháo 6 kẹp giữ nắp và mở nắp để cho hàng vào.

3 Cách sử dụng

(1) Tải trước và chiếu sáng
Tháo nắp và đổ đầy một nửa phễu chứa củi. Châm lửa vào củi từ phía dưới, đảm bảo van được mở hoàn toàn.

(2) Tải trọng đầy đủ và niêm phong
Khi gỗ bắt đầu cháy xém rõ rệt, hãy cho phần gỗ còn lại vào đầy phễu, đợi 5 phút rồi đậy kín nắp.

(3) Chiếu sáng bằng khí đốt
Khi khí gas trong suốt bắt đầu thoát ra từ ống dẫn trung tâm, hãy dùng bật lửa châm vào để bật bếp.

(4) Quy trình chạy
Nếu khói dày đặc thoát ra quá nhiều, hãy giảm lưu lượng khí. Nếu lượng khí sinh ra ít nhưng khói không nhiều, hãy tăng lưu lượng khí.

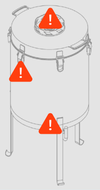
4 Thông tin kỹ thuật

(1) Khuyến nghị về gỗ
Thiết bị có thể sử dụng gỗ vụn hoặc cành cây khô, mới cắt hoặc ướt làm nhiên liệu. Kích thước khuyến nghị là từ 20mm đến 60mm (0,8" đến 2,4").

Mặc dù thiết bị được thiết kế để sử dụng với củi ướt, nhưng nên thử đốt ở giai đoạn 1 với củi khô hơn để khởi động nhanh hơn.

(2) Vệ sinh và bảo trì
Vệ sinh:
(A) Loại bỏ hết vụn gỗ hoặc mẩu gỗ lớn còn sót lại bên trong phễu.
(B) Lắc thiết bị để tro vụn rơi vào khay hứng tro.
(C) Đổ tro thu được bên dưới vào thùng ủ phân hoặc thùng rác thông thường.

Bảo trì:
(A) Gioăng và vòng chữ O có thể bị hỏng theo thời gian do hao mòn. Để thay thế, hãy tháo gioăng bị hỏng, làm sạch bề mặt và đặt gioăng mới vào.
(B) Các kẹp được lắp bằng phương pháp ép chặt, vì vậy nếu chúng bị hỏng, có thể dễ dàng lắp đặt các kẹp mới.
(C) Tay cầm bằng silicon có thể được lắp ráp bằng cách ấn vào.



(3) Thông tin an toàn
(!) Nắp, mặt bếp và bề mặt đáy của thiết bị có thể đạt đến nhiệt độ rất cao và gây bỏng cho người sử dụng.

Khi di chuyển hoặc sử dụng thiết bị, vui lòng sử dụng tay cầm bằng silicon để giảm nguy cơ bị thương.

(!) Trước khi vệ sinh thiết bị, hãy tháo nắp sau khi sử dụng và đợi 2 giờ để đảm bảo kim loại bên trong đã nguội đủ..

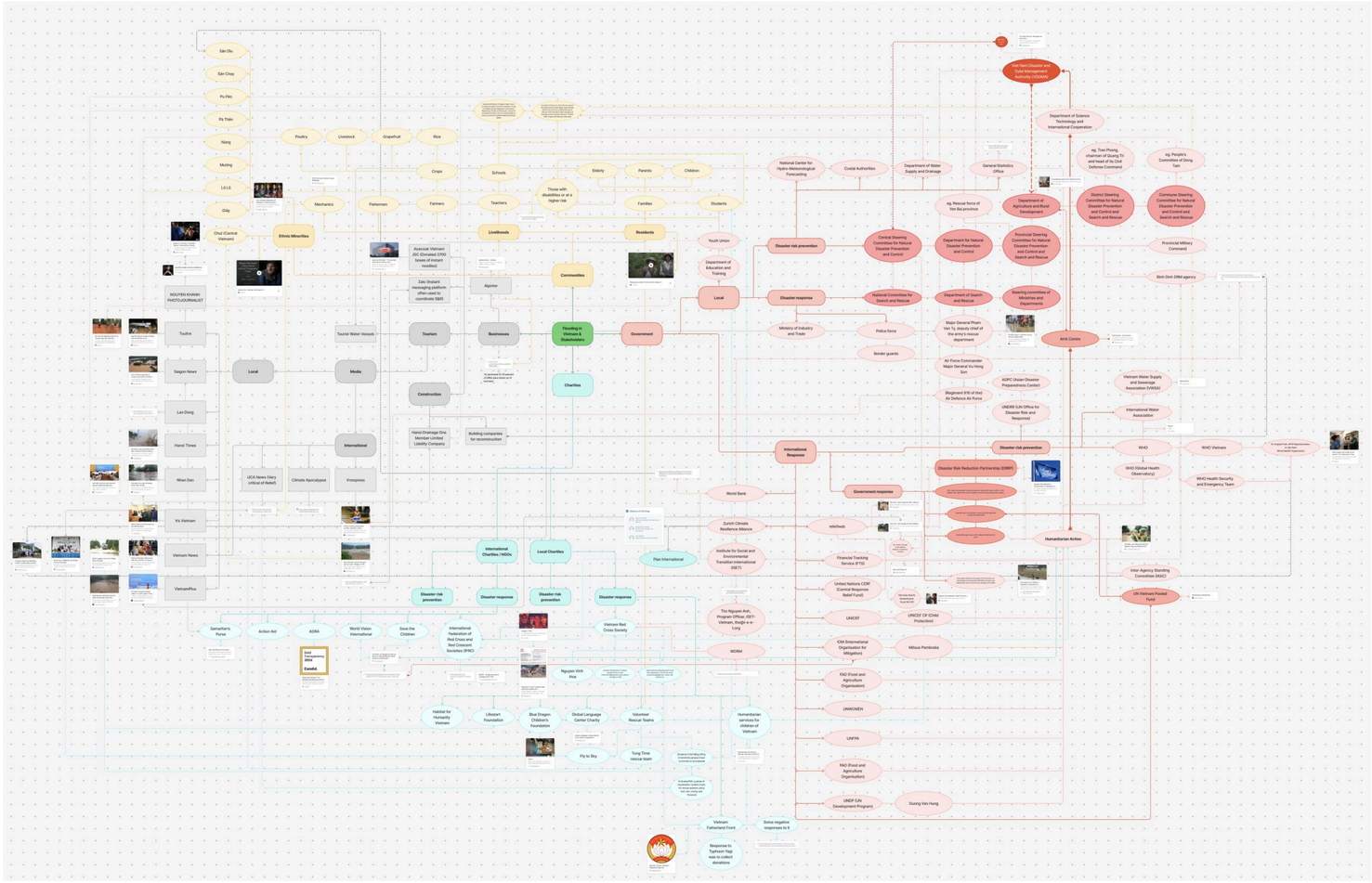
(!) Không sử dụng thiết bị bên trong.

1. The manufacture and testing of the updated design should be conducted to determine its viability and usability, and to identify any other changes that should be made.
2. The size of the wood chips should be tested in the updated design, ranging from 10mm to 60mm, to determine what works best for the device
3. The final thing that should be tested is the HAP exposure, specifically the PM2.5 and CO produced, and if there are any ways to improve it, such as by cold filtering the gas.
4. Deployment of the device should also be explored, not just in Viet Nam but also in other flood-prone areas whose populations rely heavily on firewood.

AN AMBITIOUS FIRST ATTEMPT TO PLACE A FIRST STEP INTO THE WORLD OF HUMANITARIAN AID.

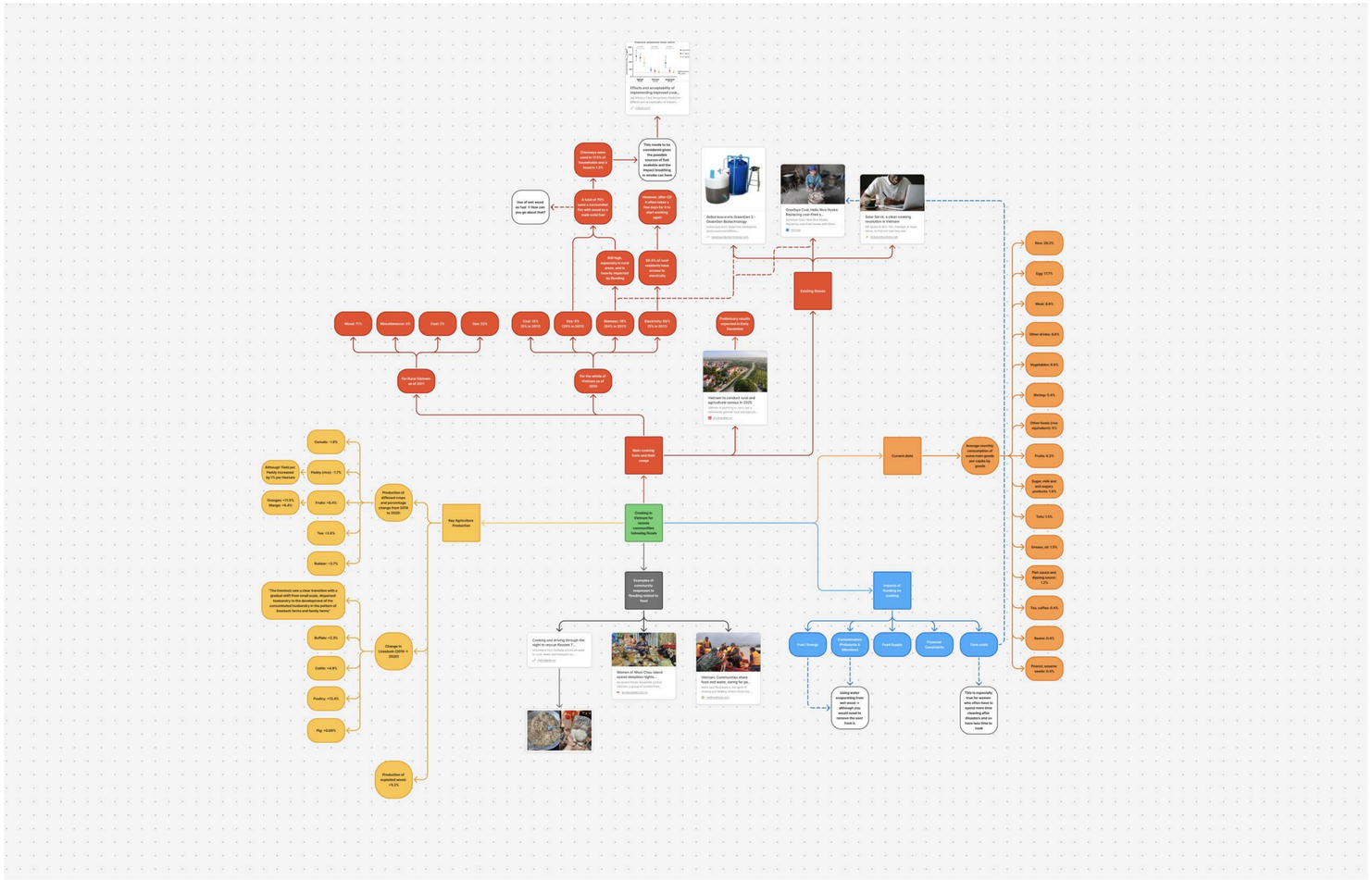
QUESTIONS AND APPENDIX

STAKEHOLDER MAP



- Rastelli F (2025) *Stakeholder map.* https://www.figma.com/board/8ppfCoVLN8RLMt93EQ5lYA/Landscape-Map?node-id=0-1&t=lrc8W6jINPvl3dXs-1 [Accessed 14th June 2026]

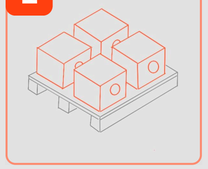
BRAINSTORMING MAP



- Rastelli F (2025) *Stakeholder map.* https://www.figma.com/board/8ppfCoVLN8RLMt93EQ5lYA/Landscape-Map?node-id=0-1&t=lrc8W6jINPvl3dXs-1 [Accessed 14th June 2026]

USER JOURNEY MAP

1

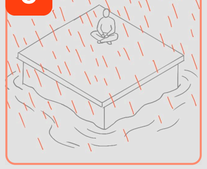


Before a flood, aid is prepared by large institutions, or individually by victims.

2

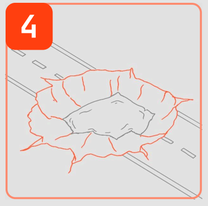
As the flood occurs, people are forced to run to the top of houses and roofs to escape.

3



Victims are often forced to sit there, exposed to the elements for days.

4



Roads are severely damage, especially in rural areas, leading to institutional aid being cut off.

5

People go along the riverbank to try and collect washed up firewood in desperation.

6

Victims are forced to help each other with no external support, sharing resources, transport and misery.

7

For some, this is the first time eating a hot meal in days. Many were found shivering from the cold.

8

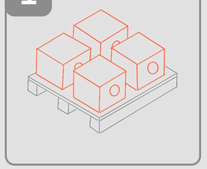
As the flood recedes, cleanup efforts begin to try and remove the mud from all surfaces.

9

In some cases, less than 2 weeks after a flood recedes, victims are hit by another disaster, re-starting the loop.

UPDATED USER JOURNEY MAP

1

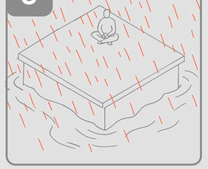


Before a flood, aid is prepared by large institutions, or individually by victims.

2

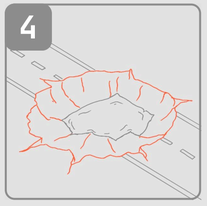
As the flood occurs, people run to the top of houses and bring with them a wet wood stove.

3



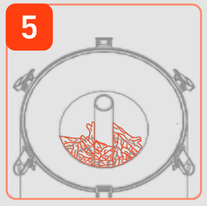
Victims are often forced to sit there, exposed to the elements for days.

4



Roads are severely damage, especially in rural areas, leading to institutional aid being cut off.

5



People, however, are able to use their firewood stores to keep warm and dry themselves.

6

Victims help each other, sharing warmth, warm meals, and firewood in a safe manner.

7

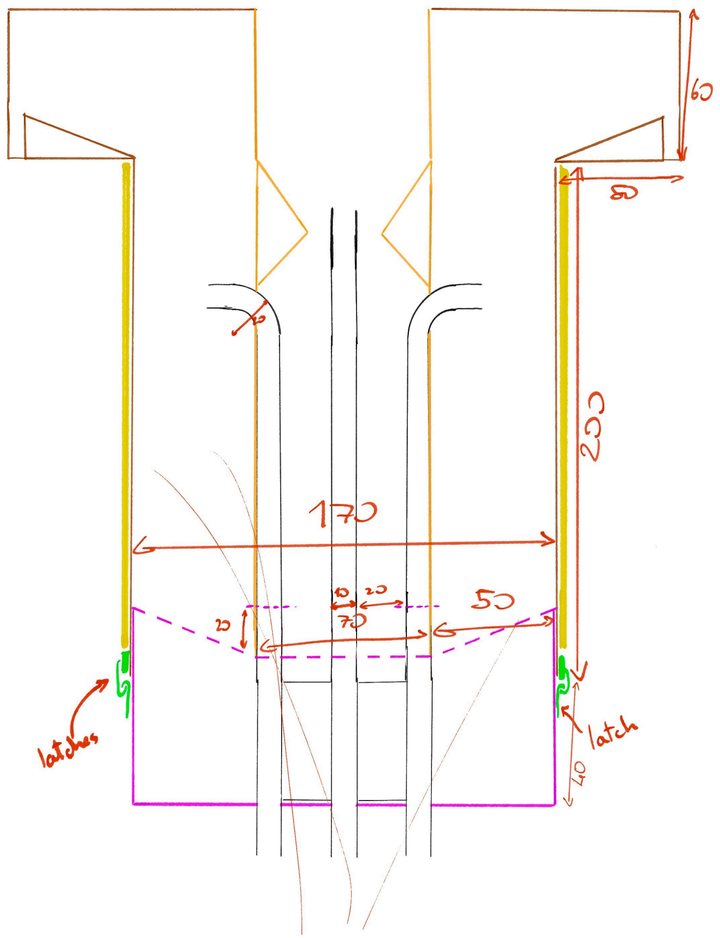
For many, they were able to eat warm food and stay warm without the need for external aid.

8

As the flood recedes, cleanup efforts begin to try and remove the mud from all surfaces.

9

In some cases, less than 2 weeks after a flood recedes, victims are hit by another disaster, re-starting the loop.



Energy to heat 750ml of water:

$$Q = m * c * \Delta T = 0.75 \times 4184 \times (100 - 24) = 238,488 J$$

Energy loss from evaporation:

$$Q = m * t * H = 37.5 \times \left(\frac{23}{60}\right) \times 2260 = 32,488\ J$$

Energy to heat 375g of rice:

$$Q = m * c * \Delta T = 0.375 \times 1800 \times (100 - 24) = 51,300 J$$

Energy to heat up a steel pot:

$$Q = m * c * \Delta T = 0.82 \times 502 \times (100 - 24) = 31,285 J$$

Total energy:

$$Q = 238,488 + 32,488 + 51,300 + 31,285 = 353,561 J$$

Total wood mass and volume:

$$M = (353,561\ \div 34,300) \div 0.35 \div 0.1 = 294.5\ g$$

$$V = 294.5 * 0.947 = \ 278.9\ cm^3$$

DISTRIBUTION ON THE DEVICE

In rural Vietnam, marketing should be reinforced by respected figures within the community, such as health workers or teachers, and focus on immediate personal benefits such as saved money or time’ (Vianello, 2016) by avoiding the need to collect dry firewood.

A combination of VLE and a Proprietary sales network would enable victims in remote rural villages to be reached and to exploit Cash & Voucher Assistance (CVA) received by the communities, facilitating easier financing for victims.

Distribution could occur both proactively and reactively.

• Vianello M. (2016) *A Review of Cooking Systems for Humanitarian Settings.* Practical Action Consulting. https://www.chathamhouse.org/sites/default/files/publications/research/2016-05-19-mei-review-of-cooking-systems-vianello.pdf

UPDRAFT VS DOWNDRAFT

- Updraft can handle up to 60% moisture content
- Has to be reloaded from the bottom which can complicate usage
- More suitable for higher-ash content as they form a simple fluidised bed

- Downdraft has 50x lower tar production due to better conditions for cracking
- Higher gas exit temperature
- Much lower HAP emissions

• Basu P (2010) Biomass gasification and pyrolysis: practical design and theory. Academic press. https://doi.org/10.1016/C2009-0-20099-7

WHY WAS MY DESIGN BLOCKED BY ASH

- Downdraft gasifier designs often have a movable bed to avoid clogging or build up on ash
- You can also have a fluidised bed where gas is pumped from the bottom shaking the ash causing it to fall through

- The original design did not incorporate either of these as I wanted to reduce the complexity to improve manufacture and reduce maintenance, and the small size of the wood chips should have helped them to crumble and fall through the holes from the weight of the chips above.

CHEMICAL REACTIONS ZONES

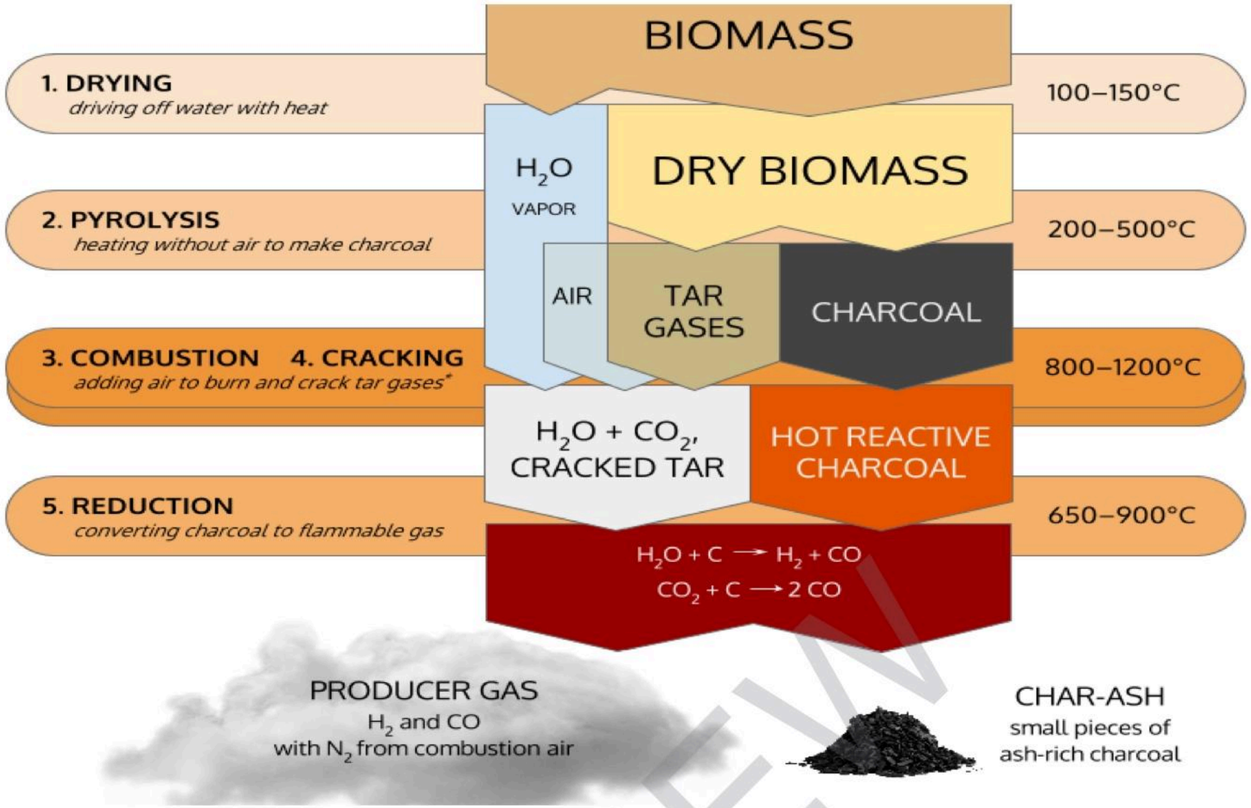


Figure 1.2. Different zones in a biomass gasification process[19]

VIEW

• Boravelli, Sai Chandra Teja, "Design, scale-up, six sigma in processing different feedstocks in a fixed bed downdraft biomass gasifier" (2016). Masters Theses. 7592. https://scholarsmine.mst.edu/masters_theses/7592

CHEMICAL REACTIONS ZONES - DRYING

- Drying: Removal of moisture.
- No chemical reaction takes place.
- This is a mass transfer operation where most of the dehydration process is due to conduction than convection.
- The weight loss by the biomass in this zone is the percentage of moisture content in the feedstock.
- H20 (l) → H2O (v)

• Boravelli, Sai Chandra Teja, "Design, scale-up, six sigma in processing different feedstocks in a fixed bed downdraft biomass gasifier" (2016). Masters Theses. 7592. https://scholarsmine.mst.edu/masters_theses/7592

CHEMICAL REACTIONS ZONES - PYROLYSIS

- Pyrolysis is a reaction where the organic material is burnt without air/ oxygen to give products such as Char, CO, H2O, and CO2 along with primary tar and secondary tar.
- These reactions take place at temperatures between 200C – 500C.
- The products causes around 85% of weight loss and the products like tar and char undergo a partial reduction in combustion and gasification zones.
- Volatile → 0.268CO + 0.295CO2 + 0.094CH4 + 0.5H2 + 0.255H2O + 0.004NH3 +0.2 primary tar
- Primary tar → 0.261secondary tar + 2.6CO 0.441CO2 + 0.983CH4 + 2.161H2 + 0.408C2H4

• Boravelli, Sai Chandra Teja, "Design, scale-up, six sigma in processing different feedstocks in a fixed bed downdraft biomass gasifier" (2016). Masters Theses. 7592. https://scholarsmine.mst.edu/masters_theses/7592

CHEMICAL REACTIONS ZONES - COMBUSTION

- Combustion: In the combustion zone, oxidation reactions take place where the carbon present in volatiles and chars reacts with air/oxygen to produce carbon dioxide.
- Hydrogen and methane react with air/oxygen to produce water vapor and carbon monoxide respectively.
- These reactions take place at temperatures ranging from 800C to 1200C.
- $C + O2 \rightarrow CO2$
- $H2 + 0.5\ O2 \rightarrow H2O$
- $CH4 + 1.5\ O2 \rightarrow CO + 2H2O$

• Boravelli, Sai Chandra Teja, "Design, scale-up, six sigma in processing different feedstocks in a fixed bed downdraft biomass gasifier" (2016). Masters Theses. 7592. https://scholarsmine.mst.edu/masters_theses/7592

CHEMICAL REACTIONS ZONES - GASIFICATION

- Reduction zone is the gasification step, where the products from oxidation steps react with red hot char present above the grate to undergo reduction.
- These reactions take place in the absence of oxygen in temperatures ranging between 650C to 900C.
- The reactions happening in this zone are as below:
  - Water gas reaction: $C + H2O \rightarrow CO + H2$
  - Water shift reaction: $CO + H2O \rightarrow CO2 + H2$
  - Boudouard reaction: $C + CO2 \rightarrow 2CO$
  - Methanation reaction: $C + 2H2 \rightarrow CH4$

• Boravelli, Sai Chandra Teja, "Design, scale-up, six sigma in processing different feedstocks in a fixed bed downdraft biomass gasifier" (2016). Masters Theses. 7592. https://scholarsmine.mst.edu/masters_theses/7592

WHY I DID NOT USE GEN-AI

“Through work, human beings bring their freedom, creativity and capacity for cooperation into play, contributing to the cultural and moral elevation of society” (Pope Leo XIV, 2026)

“Just as the creator of an artistic or literary work must consider the values it conveys, so developers are called to embed values in their projects with due seriousness: with transparency, responsibility toward affected communities and careful attention to ensuring that what is being cultivated is a genuine good.” (Pope Leo XIV, 2026)

- Pope Leo XIV (2026) *Encylical Letter Magnifica Humanitas of His Holiness Pope Leo XIV on safeguarding the human person in the time of artificial intelligence.* https://www.vatican.va/content/leo-xiv/en/encyclicals/documents/20260515-magnifica-humanitas.html [Accessed 2nd June 2026]

ORIGINAL INTENTION

Climate change is a topic which extremely interests me, but an avenue I've always struggled to go down in projects at Imperial, whether due to reasons relating to the brief, fellow teammates, or feasibility of ideas. However, I feel that this Master's project is a way for me to finally and truly enter this world, not only as a university project, but even further, as something to do after University. I feel that your guidance not only as a tutor, but also as someone who has gone and made a career out of working towards the future and climate resilience for the next 100 years is something which I would love to have as a guide for me to carve my own path into sustainable design and design for the climate as a viable career.

Having said all that, I also do not come into this project unprepared, apart from the various options which I gave you for projects I'd be interested in doing, I also have previous experience in this field having done a variety of work over the last few years:

- The most recent one was a Futures project focused around a water filtration unit for Iztapalapa, Mexico city's poorest municipality. Here we looked at using microbial fuel cells and ceramic filters to render their non-potable water safe to drink while taking into account different resources available to them: https://francescodesign.myportfolio.com/futures
- For my HCDE project, we worked with the Canal and River trust in London to find ways for them to collect rubbish quicker and reduce river pollution. While the end product ended up being software based due to the brief, I still learnt a lot from this project and even took park in volunteering work with them: https://francescodesign.myportfolio.com/hcde
- For my 6 month placement, my work focused on preparing and conducting user research, the skills I learnt here I feel are extremely helpful for me to understand the unique situation, needs and at the end of the day, the people who this product will actually be made for.

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